RAJALAKSHMI ENGINEERING COLLEGE
(AUTONOMOUS)
THANDALAM, CHENNAI 602 105

Department of Electronics and Communication Engineering



RAJALAKSHMI ENGINEERING COLLEGE

Approved by AICTE | Affiliated to Anna University | Accredited by NAAC

CS23336 Introduction to Python Programming

RECORD

NAME…… SHANTHINIDEVI.A

BRANCH…… ELECTRONICS AND COMMUNICATION ENGINEERING

REGISTER NUMBER…… 2116250801431

SEMESTER…… 3RD SEMESTER

ACADEMIC YEAR…… 2026-2027

RAJALAKSHMI ENGINEERING COLLEGE
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BONAFIDE CERTIFICATE

NAME:…SHANTHINIDEVI.A……*REGISTER NUMBER* ….2116250801431……. *Certified that this is the bonafide record of work done by the above student in the CS23336 Introduction to Python Programming during the year 2026-2027.*

Signature of the Faculty – in – charge

Submitted for the Practical Examination held on ……

INTERNAL EXAMINER **EXTERNAL EXAMINER**

RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution Affiliated to Anna University, Chennai)

DEPARTMENT OF ELECTRONICS AND

COMMUNICATION ENGINEERING REGULATIONS - 2023

VISION

To produce globally competent Electronics and Communication Engineers with a commitment to serve the society.

MISSION

M1 To impart training with the best of teaching expertise supported by excellent laboratory infrastructure and exposure to recent trends in the industry.

M2 To ensure that the students are molded into competent Electronics and Communication Engineers with the knowledge of computer applications and worthy citizens of the country.

PROGRAMME EDUCATIONAL OBJECTIVES (PEOS)

PEO I

To provide students with sound foundation in the mathematical, scientific and engineering fundamentals necessary to formulate, analyze and solve engineering problems and to prepare them for post graduate studies and for successful careers in industries.

PEO II

To develop the ability among students to define engineering problems in the fields of electronics and communication engineering, and to employ necessary techniques, hardware, and communication tools for modern engineering applications.

PEO III

To instil the values, skills, leadership and team spirit for comprehensive and wholesome personality, to promote entrepreneurial interest among students and to create a fervor for use of engineering in addressing societal concerns.

PROGRAM OUTCOMES (POs)

Engineering graduates will be able to:

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give

and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one‟s own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1: An ability to formulate solutions for practical societal requirements using communication engineering.

PSO2: To design and formulate solutions for industrial requirements using electronics and communication engineering.

PSO3: To understand and develop solutions required in multidisciplinary engineering fields.

CS23336 Introduction to Python Programming

LABORATORY

S.NO	LIST OF EXPERIMENTS	DATE	SIGNATURE
1	Experiments based on Variables, Data types and Operators in Python	03.08.2026	
2	Operators and Formatting Output	11.08.2026	
3	Algorithmic Approach: Selection control structures	18.08.3026	
4	Algorithmic Approach: Iteration control structures	29.08.2026	
5	Functions: Built-in functions, User-defined functions, Recursive functions	23.09.2026	
6	Strings and its operations	24.09.2026	
7	Lists and its operations	10.10.2026	
8	Tuples-Set and its operations	10.10.2026	
9	Dictionary and its operations	11.10.2026	
10	Linear and Binary	18.10.2026	
11	Bubble and Merge Sort	19.10.2026	
12	Files and its operations	19.10.2026	

Status	Finished
Started	Wednesday, 29 July 2026, 2:32 PM
Completed	Wednesday, 29 July 2026, 2:37 PM
Duration	5 mins 38 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which function would you use to get a Boolean value?

a. str()

b. int()

c. float()

d. bool() ✓

Your answer is correct.

The correct answer is: bool()

Question **2**

Correct

Mark 1.00 out of 1.00

Which function is used to print a message in Python?

a. echo()

b. show()

c. print() ✓

d. display()

Your answer is correct.

The correct answer is: print()

Question **3**

Correct

Mark 1.00 out of 1.00

What will be the output of the following code snippet?

print(type(5 / 2))

a. int

b. str

c. float ✓

d. obj

Your answer is correct.

The correct answer is:
float

Question **4**

Correct

Mark 1.00 out of 1.00

What will be the output of the following python Code-

mystring="India is my country"

print(type(mystring))

a. <class 'str'> ✓

b. class str

c. str

d. 'str'

Your answer is correct.

The correct answer is:
<class 'str'>

Question **5**

Correct

Mark 1.00 out of 1.00

Which Python method can be used to format strings?

a. format_string()

b. string_format()

c. str_format()

d. format() ✓

Your answer is correct.

The correct answer is: format()

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following functions is a built-in function in python language?

a. val()

b. scanf()

c. print() ✓

d. printf()

Your answer is correct.

The correct answer is:
print()

Question **7**

Correct

Mark 1.00 out of 1.00

What is the default separator between arguments in the print() function?

a. Period

b. Space ✓

c. Dash

d. Comma

Your answer is correct.

The correct answer is: Space

Question **8**

Correct

Mark 1.00 out of 1.00

What will be the datatype of the var in the below code snippet?

```
var = 10
print(type(var))
var = "Hello"
print(type(var))
```

a. No output

b. int and str ✓

c. float and str

d. int and int

Your answer is correct.

The correct answer is:
int and str

Question **9**

Incorrect

Mark 0.00 out of 1.00

Which of the following is NOT a valid identifier in Python?

a. a1

b. a_ ✗

c. 1a

d. _

Your answer is incorrect.

The correct answer is: 1a

Question **10**

Correct

Mark 1.00 out of 1.00

Which keyword is used to write comments in Python?

a. # ✓

b. /*

c. //

d. <!--

Your answer is correct.

The correct answer is: #

Question **11**

Correct

Mark 1.00 out of 1.00

What will be the output of the following Python code?

```
name = "vijay"
age = 31
print("My name is {} and I am {} years old.".format(name, age))
```

a. My name is "Vijay" and I am "31" years old.

b. My name is name and I am age years old.

c. My name is Vijay and I am 31 years old. ✓

d. My name is {} and I am {} years old.

Your answer is correct.

The correct answer is: My name is Vijay and I am 31 years old.

Question **12**

Correct

Mark 1.00 out of 1.00

What symbol is used for string formatting in Python using the old method?

a. $

b. % ✓

c. &

d. @

Your answer is correct.

The correct answer is: %

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following is a correct syntax to output “Hello Python” in Python?

a. echo("Hello Python")

b. print "Hello Python"

c. print("Hello Python") ✓

d. echo "Hello Python"

Your answer is correct.

The correct answer is: print("Hello Python")

Question **14**

Correct

Mark 1.00 out of 1.00

Which function returns a floating point number?

a. str()

b. int()

c. float() ✓

d. bool()

Your answer is correct.

The correct answer is: float()

Question **15**

Correct

Mark 1.00 out of 1.00

What will be the output of type("hello")?

a. <class 'bool'>

b. <class 'str'> ✓

c. <class 'float'>

d. <class 'int'>

Your answer is correct.

The correct answer is: <class 'str'>

Status	Finished
Started	Monday, 3 August 2026, 1:30 PM
Completed	Monday, 3 August 2026, 9:25 PM
Duration	7 hours 55 mins
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In a Logistic the Parcels to be delivered in 4 locations (1st locaion 20%, 2nd location 40%, 3rd location 30% and 4th location 10%). write a python code to find the total no. of parcels after the delivery in 2 locations . use a format() to print the no of parcels delivered in in each location

Input:

250

output:

Total Parcels is 250

1st Location 50 parcels

2nd Location 100 parcels

3rd Location 75 parcels

4th Location 25 parcels

For example:

Input	Result
250	Total Parcels is 250 1st Location 50 parcels 2nd Location 100 parcels 3rd Location 75 parcels 4th Location 25 parcels
400	Total Parcels is 400 1st Location 80 parcels 2nd Location 160 parcels 3rd Location 120 parcels 4th Location 40 parcels

Answer: (penalty regime: 0 %)

```
a=int(input())
l1=int(a*0.2)
l2=int(a*0.4)
l3=int(a*0.3)
l4=int(a*0.1)
print("Total Parcels is",a)
print(f"1st Location {l1} parcels")
print(f"2nd Location {l2} parcels")
print(f"3rd Location {l3} parcels")
print(f"4th Location {l4} parcels")

```

	Input	Expected	Got	
✓	250	Total Parcels is 250 1st Location 50 parcels 2nd Location 100 parcels 3rd Location 75 parcels 4th Location 25 parcels	Total Parcels is 250 1st Location 50 parcels 2nd Location 100 parcels 3rd Location 75 parcels 4th Location 25 parcels	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

Write a program to convert strings to an integer and float and display its type.

Sample Input:

10

10.9

Sample Output:

10,<class 'int'>

10.9,<class 'float'>

For example:

Input	Result
10 10.9	10,<class 'int'> 10.9,<class 'float'>

Answer: (penalty regime: 0 %)

```
a=int(input())
b=float(input())
print(a, type(a),sep=",")
print(round(b, 1), type(b), sep=",")
```

	Input	Expected	Got	
✓	10 10.9	10,<class 'int'> 10.9,<class 'float'>	10,<class 'int'> 10.9,<class 'float'>	✓
✓	12 12.5	12,<class 'int'> 12.5,<class 'float'>	12,<class 'int'> 12.5,<class 'float'>	✓
✓	89 7.56	89,<class 'int'> 7.6,<class 'float'>	89,<class 'int'> 7.6,<class 'float'>	✓
✓	55000 56.2	55000,<class 'int'> 56.2,<class 'float'>	55000,<class 'int'> 56.2,<class 'float'>	✓

	Input	Expected	Got	
✓	2541 2541.679	2541,<class 'int'> 2541.7,<class 'float'>	2541,<class 'int'> 2541.7,<class 'float'>	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00

Alfred buys an old scooter for Rs. X and spends Rs. Y on its repairs. If he sells the scooter for Rs. Z (Z>X+Y). Write a program to help Alfred to find his gain percent. Get all the above-mentioned values through the keyboard and find the gain percent.

Input Format:

The first line contains the Rs X

The second line contains Rs Y

The third line contains Rs Z

Sample Input:

10000

250

15000

Sample Output:

46.34 is the gain percent.

For example:

Input	Result
45500 500 60000	30.43 is the gain percent.

Answer: (penalty regime: 0 %)

```
x=int(input())
y=int(input())
z=int(input())
p=z-(x+y)
gain=(p/(x+y))*100
print(f"{gain:.2f} is the gain percent.")
```

	Input	Expected	Got	
✓	10000 250 15000	46.34 is the gain percent.	46.34 is the gain percent.	✓

	Input	Expected	Got	
✓	45500 500 60000	30.43 is the gain percent.	30.43 is the gain percent.	✓
✓	5000 0 7000	40.00 is the gain percent.	40.00 is the gain percent.	✓
✓	12500 5000 18000	2.86 is the gain percent.	2.86 is the gain percent.	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **4**

Correct

Mark 1.00 out of 1.00

Ramesh's basic salary is input through the keyboard. His dearness allowance is 40% of his basic salary, and his house rent allowance is 20% of his basic salary. Write a program to calculate his gross salary.

Sample Input:

10000

Sample Output:

16000

For example:

Input	Result
10000	16000

Answer: (penalty regime: 0 %)

```
a=int(input())
d=int(a*0.40)
h=int(a*0.20)
g=a+d+h
print(g)
```

	Input	Expected	Got	
✓	10000	16000	16000	✓
✓	20000	32000	32000	✓
✓	28000	44800	44800	✓
✓	5000	8000	8000	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **5**

Correct

Mark 1.00 out of 1.00

In department 54% are boys and 46% are girls and 8% are hostel (boys/girls). write a python code to print total no of boys, girls and hostel students in the specific format using modulo operator.

input: 1500

output: Total Students : 1500, Boys : 810, Girls : 690, Hostel : 120

For example:

Input	Result
1500	Total Students : 1500, Boys : 810, Girls : 690, Hostel : 120
1200	Total Students : 1200, Boys : 648, Girls : 552, Hostel : 96

Answer: (penalty regime: 0 %)

```
total=int(input())
boys=int(total*0.54)
girls=int(total*0.46)
hostel=int(total*0.08)
print(f"Total Students : {total}, Boys : {boys}, Girls : {girls}, Hostel : {hostel}")
```

	Input	Expected	Got	
✓	1500	Total Students : 1500, Boys : 810, Girls : 690, Hostel : 120	Total Students : 1500, Boys : 810, Girls : 690, Hostel : 120	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **6**

Correct

Mark 1.00 out of 1.00

Write a program that returns the second last digit of the given number. Second last digit is being referred 10the digit in the tens place in the given number.

For example, if the given number is 197, the second last digit is 9.

Note1 - The second last digit should be returned as a positive number. i.e. if the given number is -197, the second last digit is 9.

Note2 - If the given number is a single digit number, then the second last digit does not exist. In such cases, the program should return -1. i.e. if the given number is 5, the second last digit should be returned as -1

For example:

Input	Result
197	9
-197	9
5	-1

Answer: (penalty regime: 0 %)

```
a=int(input())
if -9<=a<=9:
    print(-1)
else:
    a=abs(a)
    b=((a%100)//10)
    print(b)
```

	Input	Expected	Got	
✓	197	9	9	✓
✓	-197	9	9	✓
✓	5	-1	-1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **7**

Correct

Mark 1.00 out of 1.00

Justin is a carpenter who works on an hourly basis. He works in a company where he is paid Rs 50 for an hour on weekdays and Rs 80 for an hour on weekends. He works 10 hrs more on weekdays than weekends. If the salary paid for him is given, write a program to find the number of hours he has worked on weekdays and weekends.

Hint:

If the final result(hrs) are in -ve convert that to +ve using abs() function

The `abs()` function returns the absolute value of the given number.

```
number = -20
absolute_number = abs(number)
print(absolute_number)
# Output: 20
```

Sample Input:

450

Sample Output:

weekdays 10.38

weekend 0.38

For example:

Input	Result
450	weekdays 10.38 weekend 0.38

Answer: (penalty regime: 0 %)

```
a=int(input())
weekend=abs((a-500)/130)
weekdays=abs(weekend+10)
print(f"weekdays {weekdays:.2f}")
print(f"weekend {weekend:.2f}")
```

	Input	Expected	Got	
✓	450	weekdays 10.38 weekend 0.38	weekdays 10.38 weekend 0.38	✓
✓	500	weekdays 10.00 weekend 0.00	weekdays 10.00 weekend 0.00	✓
✓	10000	weekdays 83.08 weekend 73.08	weekdays 83.08 weekend 73.08	✓
✓	6789	weekdays 58.38 weekend 48.38	weekdays 58.38 weekend 48.38	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **8**

Correct

Mark 1.00 out of 1.00

In a Lab 36% are Dell and 34% Lennovo and 28% are Acer and 2% are Samsung. write a python code to print total systems and brand wise count in the specific format using sep operator.

input: 150

output: Total System:150

Dell:54

Lennovo:51

Acer:42

Samsung:3

For example:

Input	Result
150	Total System:150 Dell:54 Lennovo:51 Acer:42 Samsung:3
250	Total System:250 Dell:90 Lennovo:85 Acer:70 Samsung:5

Answer: (penalty regime: 0 %)

```
a=int(input())
d=int(a*0.36)
l=int(a*0.34)
ar=int(a*0.28)
s=int(a*0.02)
print(f"Total System:{a}")
print(f"Dell:{d}")
print(f"Lennovo:{l}")
print(f"Acer:{ar}")
print(f"Samsung:{s}")
```

	Input	Expected	Got	
✓	150	Total System:150 Dell:54 Lennovo:51 Acer:42 Samsung:3	Total System:150 Dell:54 Lennovo:51 Acer:42 Samsung:3	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **9**

Correct

Mark 1.00 out of 1.00

You went on a tour to Ooty with your friends. As a part of the tour, you went boating with them. For the boat to remain stable, the number of people on one boat is restricted based on the weight of the people. You find that the boatman who is sailing your boat is so much greedy of money. For earning more, he takes too many people to travel in the boat at a time. So you want to check how many people can travel in the boat at a time so that the boat will not drown. Calculate the weight by considering the number of adults and number of children. Assume that an adult weighs 75 kg and children weigh 30 kg each. If the weight is normal, display Boat is stable, else display Boat will drown.

INPUT & OUTPUT FORMAT:

Input consists of 3 integers.

First input corresponds to the weight that the boat can handle.

Second input corresponds to the number of adults.

Third input corresponds to the number of children.

For example:

Input	Result
340 2 3	Boat is stable
600 7 4	Boat will drow

Answer: (penalty regime: 0 %)

```
a=int(input())
b=int(input())
c=int(input())
d=b*75
e=c*30
f=d+e
if(a>f):
    print("Boat is stable")
else:
    print("Boat will drow")
```

	Input	Expected	Got	
✓	340 2 3	Boat is stable	Boat is stable	✓

	Input	Expected	Got	
✓	600 7 4	Boat will drow	Boat will drow	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **10**

Correct

Mark 1.00 out of 1.00

In many jurisdictions, a small deposit is added to drink containers to encourage people to recycle them. In one particular jurisdiction, drink containers holding one liter or less have a $0.10 deposit and drink containers holding more than one liter have a $0.25 deposit. Write a program that reads the number of containers of each size(less and more) from the user. Your program should continue by computing and displaying the refund that will be received for returning those containers. Format the output so that it includes a dollar sign and always displays exactly two decimal places.

Sample Input

10

20

Sample Output

Your total refund will be $6.00.

For example:

Input	Result
20 20	Your total refund will be $7.00.

Answer: (penalty regime: 0 %)

```
a=int(input())
b=int(input())
c=a*0.10
d=b*0.25
total=c+d
print(f"Your total refund will be ${total:.2f}.")
```

	Input	Expected	Got	
✓	20 20	Your total refund will be $7.00.	Your total refund will be $7.00.	✓
✓	11 22	Your total refund will be $6.60.	Your total refund will be $6.60.	✓
✓	123 200	Your total refund will be $62.30.	Your total refund will be $62.30.	✓

	Input	Expected	Got	
✓	76 38	Your total refund will be $17.10.	Your total refund will be $17.10.	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Status	Finished
Started	Sunday, 9 August 2026, 3:16 PM
Completed	Sunday, 9 August 2026, 3:25 PM
Duration	8 mins 11 secs
Grade	**14.33** out of 15.00 (**95.56**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the order of precedence in python?

1. **Multiplication**
2. **Division**
3. **Parentheses**
4. **Addition**
5. **Exponentiation**

a. **3,1,2,4,5**

b. **1,5,2,4,3**

3,1,2,4,5

c. **1,2,3,4,5**

d. **3,5,1,2,4**

The correct answer is:

3,5,1,2,4

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following is not a valid variable name in Python?

a. **_var**

b. **var11**

c. **5var**

d. **var_name**

The correct answer is:

5var

Correct

Marks for this submission: 1.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00

What is the value of the expression

```
print(100 / 25)
print(100//25)
```

a. 4.0
4.00

b. 4.0
4

c. 4
4

d. 4.0
4.0

The correct answer is:
4.0
4

Correct

Marks for this submission: 1.00/1.00.

Question **4**

Correct

Mark 1.00 out of 1.00

An identifier can have a maximum length of -------- characters in Python.

a. 31

b. 50

c. 7

d. 79

The correct answer is:
79

Correct

Marks for this submission: 1.00/1.00.

Question **5**

Correct

Mark 1.00 out of 1.00

What is the output of the following assignment operator?

```
y = 10
x = y += 2
print(x)
```

a. Syntax Error

b. 10

c. 12

d. 14

The correct answer is:
Syntax Error

Correct

Marks for this submission: 1.00/1.00.

Question **6**

Correct

Mark 1.00 out of 1.00

What will be the value of x in the following Python expression, if the result of that expression is 2?

x>>2

a. 4

b. 2

c. 1

d. 8

The correct answer is:
8

Correct

Marks for this submission: 1.00/1.00.

Question **7**

Correct

Mark 1.00 out of 1.00

What is the output of the following code

```
x = 4
y = 10

print(x % y)
```

a. 1

b. 6

c. 10

d. 4

The correct answer is:
4

Correct

Marks for this submission: 1.00/1.00.

Question **8**

Correct

Mark 1.00 out of 1.00

What is the two's complement of -44?

a. 11101011

b. 10110011

c. 1011011

d. 11010100

The correct answer is:
11010100

Correct

Marks for this submission: 1.00/1.00.

Question **9**

Correct

Mark 1.00 out of 1.00

State the output of the following code.

```
num1 = '10'
num2 = '20'
sum = num1 + num2
print(sum)
```

a. Error

b. 10

c. 1020

d. 30

The correct answer is:
1020

Correct

Marks for this submission: 1.00/1.00.

Question **10**

Correct

Mark 1.00 out of 1.00

What is the output of the following code

```
x = 8
y = 2
print(x ** y)
print(x // y)
```

a. 64
8
4

b. 0
64

c. 64
4

d. 64
0

The correct answer is:
64
4

Correct

Marks for this submission: 1.00/1.00.

Question **11**

Correct

Mark 1.00 out of 1.00

What is the output of the following code

```
x = ["apple", "banana"]
y = ["apple", "banana"]
z = x
 print(x is z)
print(x is y)
print(x == y)
```

a. **True**
False
False

b. **True**
False
True

c. **False**
False
True

d. **True**
True
True

The correct answer is:
True
False
True

Correct

Marks for this submission: 1.00/1.00.

Question **12**

Correct

Mark 1.00 out of 1.00

What is the output of the following code

```
x = 1j
 #display x:
print(x)
 #display the data type of x:
print(type(x))
```

a. **1j**
<class 'complex'>

b. **1**
<class 'int'>

c. **1j.0**
<class 'float'>

d. **1j**
<class 'object'>

The correct answer is:
1j
<class 'complex'>

Correct

Marks for this submission: 1.00/1.00.

Question **13**

Correct

Mark 1.00 out of 1.00

What will be the output of statement 22**2**2**

a. 65536

b. 32768

c. 256

d. 16

The correct answer is:
65536

Correct

Marks for this submission: 1.00/1.00.

Question **14**

Correct

Mark 0.33 out of 1.00

What is the value of the expression `1+2**3*4+12*((100+4)*10-200//10)` ?

a. -24568

b. 12493

c. -23679

d. 12273

The correct answer is:

12273

Correct

Marks for this submission: 1.00/1.00. Accounting for previous tries, this gives **0.33/1.00**.

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following type of Python operator will only print True or False in output when we use it in our program?

a. Assignment Operator

b. Comparison Operator

c. Membership Operator

d. Arithmetic Operator

The correct answers are:

Membership Operator,

Comparison Operator

Correct

Marks for this submission: 1.00/1.00.

Status	Finished
Started	Sunday, 9 August 2026, 3:28 PM
Completed	Tuesday, 11 August 2026, 2:32 PM
Duration	1 day 23 hours
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Write a program that returns the last digit of the given number. Last digit is being referred to the least significant digit i.e. the digit in the ones (units) place in the given number.

The last digit should be returned as a positive number.

For example,

if the given number is 197, the last digit is 7

if the given number is -197, the last digit is 7

For example:

Input	Result
197	7
-197	7

Answer: (penalty regime: 0 %)

```
a=int(input())
b=abs(a)
num=((b%100)%10)
print(num)
```

	Input	Expected	Got	
✓	197	7	7	✓
✓	-197	7	7	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

In the 1800s, the battle of Troy was led by Hercules. He was a superstitious person. He believed that his crew can win the battle only if the total count of the weapons in hand is in multiple of 3 and the soldiers are in an even number of count. Given the total number of weapons and the soldier's count, Find whether the battle can be won or not according to Hercules's belief. If the battle can be won print True otherwise print False.

Input format:

Line 1 has the total number of weapons

Line 2 has the total number of Soldiers.

Output Format:

If the battle can be won print True otherwise print False.

Sample Input:

32

43

Sample Output:'

False

For example:

Input	Result
32 43	False

Answer: (penalty regime: 0 %)

```
weapons=int(input())
soldiers=int(input())
if(weapons%3==0 & soldiers%2==0):
    print("True")
else:
    print("False")
```

	Input	Expected	Got	
✓	32 43	False	False	✓
✓	273 7890	True	True	✓
✓	800 4590	False	False	✓
✓	6789 32996	True	True	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00

Write a simple python program to find the square root of a given floating point number. The output should be displayed with 3 decimal places.

Sample Input:

8.00

Sample Output:

2.828

For example:

Input	Result
8.00	2.828
14.00	3.742

Answer: (penalty regime: 0 %)

```
a=float(input())
square=a**(1/2)
print(f"{square:.3f}")
```

	Input	Expected	Got	
✓	8.00	2.828	2.828	✓
✓	14.00	3.742	3.742	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **4**

Correct

Mark 1.00 out of 1.00

Mr.Ram has been given a problem kindly help him to solve it. The input of the program is either 0 or 1. IF 0 is the input he should display "C" if 1 is the input it should display "D".There is a constraint that Mr. Ram should use either logical operators or arithmetic operators to solve the problem, not anything else.

Hint:

Use ASCII values of C and D.

Input Format:

An integer x, 0<=x<=1. .

Output Format:

output a single character "C" or "D"depending on the value of x.

```
Input 1:
0
Output 1:
C
```

```
Input 2:
1

Output 1:
D
```

For example:

Input	Result
0	C

Answer: (penalty regime: 0 %)

```
x=int(input())
if x==0:
    print("C")
else:
    print("D")
```

	Input	Expected	Got	
✓	0	C	C	✓
✓	1	D	D	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **5**

Correct

Mark 1.00 out of 1.00

Write a program to find whether the given input number is Even.

If the given number is even, the function should return 2 else it should return 1.

Note: The number passed to the program can either be negative, positive or zero. Zero should be treated as Even.

For example:

Input	Result
100	2
1001	1

Answer: (penalty regime: 0 %)

```
num=int(input())
if(num%2==0):
    print("2")
else:
    print("1")
```

	Input	Expected	Got	
✓	100	2	2	✓
✓	1001	1	1	✓
✓	0	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **6**

Correct

Mark 1.00 out of 1.00

Rohit wants to add the last digits of two given numbers.

For example,

If the given numbers are 267 and 154, the output should be 11.

Below is the explanation:

Last digit of the 267 is 7

Last digit of the 154 is 4

Sum of 7 and 4 = 11

Write a program to help Rohit achieve this for any given two numbers.

Note: Tile sign of the input numbers should be ignored.

i.e.

if the input numbers are 267 and 154, the sum of last two digits should be 11

if the input numbers are 267 and -154, the slim of last two digits should be 11

if the input numbers are -267 and 154, the sum of last two digits should be 11

if the input numbers are -267 and -154, the sum of last two digits should be 11

For example:

Input	Result
267 154	11
267 -154	11

Answer: (penalty regime: 0 %)

```
a=int(input())
b=int(input())
c=abs(a)
d=abs(b)
e=((c%100)%10)
f=((d%100)%10)
g=e+f
print(g)
```

	Input	Expected	Got	
✓	267 154	11	11	✓
✓	267 -154	11	11	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **7**

Correct

Mark 1.00 out of 1.00

Complete the program to convert days into years, month and days. (Ignoring leap year and considering 1 month is 30 days)

Sample Test Cases

Test Case 1

Input

375

Output

YEARS: 1 MONTH: 0 DAYS: 10

Test Case 2

Input

200

Output

YEARS: 0 MONTH: 6 DAYS: 20

Answer: (penalty regime: 0 %)

```
days=int(input())
years=days//365
remaining_days=days%365
months=remaining_days//30
days_r=remaining_days%30
print(f"YEARS: {years} MONTH: {months} DAYS: {days_r}")
```

	Input	Expected	Got	
✓	375	YEARS: 1 MONTH: 0 DAYS: 10	YEARS: 1 MONTH: 0 DAYS: 10	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **8**

Correct

Mark 1.00 out of 1.00

An online retailer sells two products: widgets and gizmos. Each widget weighs 75 grams. Each gizmo weighs 112 grams. Write a program that reads the number of widgets and the number of gizmos from the user. Then your program should compute and display the total weight of the parts.

Sample Input:

10

20

Sample Output:

The total weight of all these widgets and gizmos is 2990 grams.

Answer: (penalty regime: 0 %)

```
a=int(input())
b=int(input())
widget=a*75
gizmo=b*112
total_weight=widget+gizmo
print(f"The total weight of all these widgets and gizmos is {total_weight} grams.")
```

	Input	Expected	Got	
✓	10 20	The total weight of all these widgets and gizmos is 2990 grams.	The total weight of all these widgets and gizmos is 2990 grams.	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **9**

Correct

Mark 1.00 out of 1.00

Write a python program that takes a integer between 0 and 15 as input and displays the number of '1' s in its binary form.(Hint:use python bitwise operator.

Sample Input

3

Sample Output:

2

Explanation:

The binary representation of 3 is 011, hence there are 2 ones in it. so the output is 2.

For example:

Input	Result
3	2

Answer: (penalty regime: 0 %)

```
a=int(input())
binary=bin(a)
ones=binary.count("1")
print(ones)
```

	Input	Expected	Got	
✓	3	2	2	✓
✓	5	2	2	✓
✓	15	4	4	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **10**

Correct

Mark 1.00 out of 1.00

Note:

Dont use if-else. Operators alone must be used .

A team from the Rotract club had planned to conduct a rally to create awareness among the Coimbatore people to donate blood. They conducted the rally successfully. Many of the Coimbatore people realized it and came forward to donate their blood to nearby blood banks. The eligibility criteria for donating blood are people should be above or equal to 18 and his/ her weight should be above 40. There was a huge crowd and staff in the blood bank found it difficult to manage the crowd. So they decided to keep a system and ask the people to enter their age and weight in the system. If a person is eligible he/she will be allowed inside.

Write a program and feed it to the system to find whether a person is eligible or not.

Input Format:

Input consists of two integers that correspond to the age and weight of a person respectively.

Output Format:

Display True(IF ELIGIBLE)

Display False (if not eligible)

Sample Input

19

45

Sample Output

True

For example:

Input	Result
18 40	False

Answer: (penalty regime: 0 %)

```
a=int(input())
b=int(input())
if(a>=18 and b>40):
    print("True")
else:
    print("False")
```

	Input	Expected	Got	
✓	19 45	True	True	✓
✓	18 40	False	False	✓
✓	18 42	True	True	✓
✓	16 45	False	False	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Status	Finished
Started	Tuesday, 18 August 2026, 4:18 PM
Completed	Tuesday, 18 August 2026, 4:21 PM
Duration	2 mins 37 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What will be the output of the given code?

```
x,y=1,1
if(x ==y):
    print("equal")
if(x>y):
    print("1")
else:
    print("0")
```

a. 1

b. equal

c. 0

d. equal
0

Question **2**

Complete

Mark 1.00 out of 1.00

What is the output of the following snippet if 5 is given as the input?

```
c = input("Enter a number: ")
print(c * 3)
```

a. 15

b. 8

c. 2

d. 555

Question **3**

Complete

Mark 1.00 out of 1.00

Which of the following statements correctly represents taking input from user in python?

a. None of the mentioned

b. a=input("Enter the value")

c. a=get("Enter the value")

d. a=inp("Enter the value")

Question **4**

Complete

Mark 1.00 out of 1.00

____ is an empty statement in Python.

a. Empty

b. pass

c. None

d. Jump

Question **5**

Complete

Mark 1.00 out of 1.00

selection is implemented with the help of _____ statement

a. for loop

b. while loop

c. if..else

Question **6**

Complete

Mark 1.00 out of 1.00

Which of the following options correctly prints the phrase "Hurry Up!!"?

a. printf("Hurry Up!!")

b. print_sentence(Hurry Up!!!)

c. print("Hurry Up!!")

d. print(Hurry Up!!!)

Question **7**

Complete

Mark 1.00 out of 1.00

What is the output of the following code

```
x=3
if x>2 or x<5 and x==6:
    print("ok")
else:
    print("no output")
```

a. None of the given option

b. error

c. ok

d. no output

Question **8**

Complete

Mark 1.00 out of 1.00

To write else statement in if-elif ladder is mandatory?

a. True

b. False

Question **9**

Complete

Mark 1.00 out of 1.00

Which of the following is true about the code below?

```
x = 3
if (x > 2):
    x = x * 2;
if (x > 4):
    x = 0;
print(x)
```

a. x will always equal 0 after this code executes for any value of x

b. if x is lesser then 0,x will be 0 after this code executes

c. if x is greater than 2, x will equal 0 after this code executes

d. if x is greater than 2, the value in x will be doubled after this code executes

Question **10**

Complete

Mark 1.00 out of 1.00

What is the output of the below code snippet?

```
num1=100
num2=200
num3=6
if(5>=num3):
    if(num1>100 or num2>150):
        print("1")
elif(num1>=100 and num2>150):
    print("2")
else:
    print("3")
```

a. 2

b. 1

c. 3

Question **11**

Complete

Mark 1.00 out of 1.00

What is the output

```
a, b, c = 1, 3, 5

if a + b + c:

    print("Hello")

else:

    print("Know Program")
```

a. Compiled Successfully, No Output.

b. Error

c. Know Program

d. Hello

Question **12**

Complete

Mark 1.00 out of 1.00

What is the output when the following sequence of instructions is carried out in the console?

```
a = 1;a = a + 1;a = a + 2;a = a + 3;print(a)
```

a. 7

b. 6

c. 5

d. 4

Question **13**

Complete

Mark 1.00 out of 1.00

Python supports _____ types of control structures.

a. 1
b. 2
c. 3
d. 4

Question **14**

Complete

Mark 1.00 out of 1.00

Which of the following statements in Python is used when no operation is to be performed?

a. pass
b. None
c. Jump
d. Empty

Question **15**

Complete

Mark 1.00 out of 1.00

What keyword would you use to add an alternative condition to an if statement?

a. else if
b. elif
c. None of the above
d. elseif

Status	Finished
Started	Monday, 17 August 2026, 1:52 PM
Completed	Tuesday, 18 August 2026, 3:47 PM
Duration	1 day 1 hour
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Write a program to calculate and print the Electricity bill where the unit consumed by the user is given from test case. It prints the total amount the customer has to pay. The charge are as follows:

Unit	Charge / Unit
Upto 199	@1.20
200 and above but less than 400	@1.50
400 and above but less than 600	@1.80
600 and above	@2.00

If bill exceeds Rs.400 then a surcharge of 15% will be charged and the minimum bill should be of Rs.100/-

Sample Test Cases

Test Case 1

Input

50

Output

100.00

Test Case 2

Input

300

Output

517.50

For example:

Input	Result
100.00	120.00
500	1035.00

Answer: (penalty regime: 0 %)

```
a=float(input())
if a<=199:
    b=a*1.20
elif 200<=a<=400:
    b=a*1.50
elif 400<=a<=600:
    b=a*1.80
elif a>600:
    b=a*2.00
if b>=400:
    b=b*1.15
elif b<=100:
    b=100
print(f"{b:.2f}")
```

	Input	Expected	Got	
✓	50	100.00	100.00	✓
✓	100.00	120.00	120.00	✓
✓	500	1035.00	1035.00	✓
✓	700	1610.00	1610.00	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

Given an integer N, check whether N the given number can be made a perfect square after adding 1 to it.

Input Format:

Single integer input.

Output Format:

Yes or No.

Example Input:

24

Output:

Yes

Example Input:

26

Output:

No

For example:

Input	Result
24	Yes

Answer: (penalty regime: 0 %)

```
a=int(input())
b=a+1
c=int(b**0.5)
if(c*c==b):
    print("Yes")
else:
    print("No")
```

	Input	Expected	Got	
✓	24	Yes	Yes	✓
✓	26	No	No	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00

Write a program to determine the type of berth when the seat / berth number in the train is given.



Input Format:

Input consists of a single integer. Assume that the range of input is between 1 and 72.

Output Format:

Output consists of a single string. [Upper or Middle or Lower or Side Lower or Side Upper]

Sample Input 1:

9

Sample Output 1:

Lower Berth

Answer: (penalty regime: 0 %)

```
i=int(input())
if i%8==1 or i%8==4:
    print("Lower Berth")
elif i%8==2 or i%8==5:
    print("Middle Berth")
elif i%8==3 or i%8==6:
    print("Upper Berth")
elif i%8==7:
    print("Side Lower Berth")
else:
    print("Side Upper Berth")
```

	Input	Expected	Got	
✓	9	Lower Berth	Lower Berth	✓
✓	72	Side Upper Berth	Side Upper Berth	✓
✓	20	Lower Berth	Lower Berth	✓
✓	34	Middle Berth	Middle Berth	✓
✓	39	Side Lower Berth	Side Lower Berth	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **4**

Correct

Mark 1.00 out of 1.00

The Chinese zodiac assigns animals to years in a 12 year cycle. One 12 year cycle is shown in the table below. The pattern repeats from there, with 2012 being another year of the dragon, and 1999 being another year of the hare.

Year Animal

2000 Dragon

2001 Snake

2002 Horse

2003 Sheep

2004 Monkey

2005 Rooster

2006 Dog

2007 Pig

2008 Rat

2009 Ox

2010 Tiger

2011 Hare

Write a program that reads a year from the user and displays the animal associated with that year. Your program should work correctly for any year greater than or equal to zero, not just the ones listed in the table.

Sample Input 1

2010

Sample Output 1

2010 is the year of the Tiger.

Sample Input 2

2020

Sample Output 2

2020 is the year of the Rat.

Answer: (penalty regime: 0 %)

```
year=int(input())
animals=["Dragon","Snake","Horse","Sheep","Monkey","Rooster","Dog","Pig","Rat","Ox","Tiger","Hare"]
rem=animals[(year-2000)%12]
print(f"{year} is the year of the {rem}.")
```

	Input	Expected	Got	
✓	2010	2010 is the year of the Tiger.	2010 is the year of the Tiger.	✓
✓	2020	2020 is the year of the Rat.	2020 is the year of the Rat.	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **5**

Correct

Mark 1.00 out of 1.00

Most years have 365 days. However, the time required for the Earth to orbit the Sun is actually slightly more than that. As a result, an extra day, February 29, is included in some years to correct for this difference. Such years are referred to as leap years. The rules for determining whether or not a year is a leap year follow:

• Any year that is divisible by 400 is a leap year.

• Of the remaining years, any year that is divisible by 100 is not a leap year.

• Of the remaining years, any year that is divisible by 4 is a leap year.

• All other years are not leap years.

Write a program that reads a year from the user and displays a message indicating whether or not it is a leap year.

Sample Input 1

1900

Sample Output 1

1900 is not a leap year.

Sample Input 2

2000

Sample Output 2

2000 is a leap year.

Answer: (penalty regime: 0 %)

```
year=int(input())
if(year%400==0):
    print(f"{year} is a leap year.")
elif(year%100==0):
    print(f"{year} is not a leap year.")
elif(year%4==0):
    print(f"{year} is a leap year.")
else:
    print(f"{year} is not a leap year.")

```

	Input	Expected	Got	
✓	1900	1900 is not a leap year.	1900 is not a leap year.	✓
✓	2000	2000 is a leap year.	2000 is a leap year.	✓
✓	2100	2100 is not a leap year.	2100 is not a leap year.	✓
✓	2020	2020 is a leap year.	2020 is a leap year.	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **6**

Correct

Mark 1.00 out of 1.00

Write a program that accepts 5 inputs and returns the count of how many of those 5 are odd.

For example,

If the five inputs are 12, 17, 19, 14, and 115, there are three odd numbers 17, 19 and 115. So, the program must return 3.

Similarly,

If the five inputs are 15, 0, -12, 19, and 28, there are two odd numbers 15 and 19. So, the program must return 2.

Observe that zero is considered an even number.

For example:

Input	Result
12 17 19 14 115	3
15 0 -12 19 28	2

Answer: (penalty regime: 0 %)

```
count=0
for i in range(5):
    n=int(input())
    if n%2 !=0:
        count+=1
print(count)
```

	Input	Expected	Got	
✓	12 17 19 14 115	3	3	✓

	Input	Expected	Got	
✓	15 0 -12 19 28	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **7**

Correct

Mark 1.00 out of 1.00

Write a Python program that accepts three parameters. The first parameter is an integer. The second is one of the following mathematical operators: +, -, /, or *. The third parameter will also be an integer.

The function should perform a calculation and return the results. For example, if the function is passed 6 and 4, it should return 24.

Sample Input Format:

11

+

14

Sample Output Format:

25

Answer: (penalty regime: 0 %)

```
a=int(input())
op=input()
b=int(input())
if (op=="+"):
    print(a+b)
elif(op=="-"):
    print(a-b)
elif(op=="*"):
    print(a*b)
elif(op=="/"):
    print(a/b)
```

	Input	Expected	Got	
✓	11 + 14	25	25	✓
✓	45 - 50	-5	-5	✓
✓	12 * 100	1200	1200	✓

	Input	Expected	Got	
✓	18 / 2	9.0	9.0	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **8**

Correct

Mark 1.00 out of 1.00

A triangle can be classified based on the lengths of its sides as equilateral, isosceles or scalene. All three sides of an equilateral triangle have the same length. An isosceles triangle has two sides that are the same length, and a third side that is a different length. If all of the sides have different lengths then the triangle is scalene.

Write a program that reads the lengths of the three sides of a triangle from the user. Then display a message that states the triangle's type.

Sample Input 1

60

60

60

Sample Output 1

That's a equilateral triangle

Sample Input 2

40

40

80

Sample Output 2

That's a isosceles triangle

Sample Input 3

50

60

70

Sample Output 3

That's a scalene triangle

For example:

Input	Result
60 60 60	That's a equilateral triangle
40 40 80	That's a isosceles triangle

Answer: (penalty regime: 0 %)

```
a=int(input())
b=int(input())
c=int(input())
if(a==b==c):
    print("That's a equilateral triangle")
elif(a==b!=c):
    print("That's a isosceles triangle")
elif(a!=b!=c):
    print("That's a scalene triangle")
```

	Input	Expected	Got	
✓	60 60 60	That's a equilateral triangle	That's a equilateral triangle	✓
✓	40 40 80	That's a isosceles triangle	That's a isosceles triangle	✓
✓	50 60 70	That's a scalene triangle	That's a scalene triangle	✓
✓	50 50 80	That's a isosceles triangle	That's a isosceles triangle	✓
✓	10 10 10	That's a equilateral triangle	That's a equilateral triangle	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **9**

Correct

Mark 1.00 out of 1.00

IN / OUT

Ms. Sita, the faculty handling programming lab for you is very strict. Your seniors have told you that she will not allow you to enter the week's lab if you have not completed atleast half the number of problems given last week. Many of you didn't understand this statement and so they requested the good programmers from your batch to write a program to find whether a student will be allowed into a week's lab given the number of problems given last week and the number of problems solved by the student in that week.

Input Format:

Input consists of 2 integers.

The first integer corresponds to the number of problems given and the second integer corresponds to the number of problems solved.

Output Format:

Output consists of the string "IN" or "OUT".

Sample Input and Output:

Input

8

3

Output

OUT

For example:

Input	Result
8 3	OUT

Answer: (penalty regime: 0 %)

```
a=int(input())
b=int(input())
if(b>=(a//2)):
    print("IN")
elif(b<=(a//2)):
    print("OUT")
```

	Input	Expected	Got	
✓	8 3	OUT	OUT	✓
✓	8 5	IN	IN	✓
✓	20 9	OUT	OUT	✓
✓	50 31	IN	IN	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **10**

Correct

Mark 1.00 out of 1.00

A certain type of steel is used to test and give grade according to the following conditions.

1. Hardness of the steel must be greater than 50
2. Carbon content of the steel must be less than 0.7
3. Tensile strength must be greater than 5600

The grades awarded are as follows:

- Grade is 10 if all three conditions are met
- Grade is 9 if conditions (1) and (2) are met
- Grade is 8 if conditions (2) and (3) are met
- Grade is 7 if conditions (1) and (3) are met
- Grade is 6 if only one condition is met
- Grade is 5 if none of the three conditions are met

Write a program to display the grade of the steel, based on the values of hardness, carbon content and tensile strength of the steel, given by the user.

Input

53

0.6

5602

Output:

10

Answer: (penalty regime: 0 %)

```
h=int(input())
c=float(input())
t=int(input())
if(h>=50 and c<=0.7 and t>=5600):
    print("10")
elif(h>=50 and c<=0.7):
    print("9")
elif(c<=0.7 and t>=5600):
    print("8")
elif(h>=50 and t>=5600):
    print("7")
elif(h>=50 or c<=0.7 or t>=5600):
    print("6")
else:
    print("5")
```

	Input	Expected	Got	
✓	53 0.6 5602	10	10	✓
✓	45 0 4500	6	6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Status	Finished
Started	Monday, 17 August 2026, 9:16 PM
Completed	Monday, 17 August 2026, 9:31 PM
Duration	14 mins 56 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

selection is implemented with the help of _____ statement

- a. for loop
- b. if..else ✓
- c. while loop

Question **2**

Correct

Mark 1.00 out of 1.00

Correct syntax of writing 'simple if' statement is _____

- a. **if (condition)**
  statements
- b. **if condition**
  statements
- c. **if condition --**
  statements
- d. **if condition :** ✓
  statements

Question **3**

Correct

Mark 1.00 out of 1.00

Given the nested `if-else` below, what will be the value `x` when the code executed successfully

```
x = 0
a = 5
b = 5
if a > 0:
    if b < 0:
        x = x + 5
    elif a > 5:
        x = x + 4
    else:
        x = x + 3
else:
    x = x + 2
print(x)
```

a. 0

b. 4

c. 2

d. 3 ✓

Question **4**

Correct

Mark 1.00 out of 1.00

What is the output of the following code

```
x=3
if x>2 or x<5 and x==6:
    print("ok")
else:
    print("no output")
```

a. no output

b. None of the given option

c. ok ✓

d. error

Question **5**

Correct

Mark 1.00 out of 1.00

What is the output of the following snippet if 5 is given as the input?

```
c = input("Enter a number: ")
print(c * 3)
```

a. 8

b. 15

c. 555 ✓

d. 2

Question **6**

Correct

Mark 1.00 out of 1.00

What is the output of the following code snippet?

```
a = "Hi"
b = "Arjuna"
c = "Bhimaa"
print("Hi", a, b, c)
```

a. Hi Hi Arjuna Bhimaa ✓

b. Hi Arjuna Hi Bhimaa

c. Hi Arjuna Bhimaa Hi

d. Hi Arjuna Bhimaa

Question **7**

Correct

Mark 1.00 out of 1.00

Write the output of the following code :

```
y=2
if 2!=y:
 print("H")
else :
 print("K")
```

a. H

b. No output

c. K ✓

d. Error

Question **8**

Correct

Mark 1.00 out of 1.00

What is the output of the given below program?

```
if 1 + 3 == 7:
    print("Hello")
else:
    print("Know Program")
```

a. Hello

b. Know Program ✓

c. Error

d. Compiled Successfully, No Output.

Question **9**

Correct

Mark 1.00 out of 1.00

What is the output

```
a, b, c = 1, 3, 5
if a + b + c:
    print("Hello")
else:
    print("Know Program")
```

a. Hello ✓

b. Error

c. Compiled Successfully, No Output.

d. Know Program

Question **10**

Correct

Mark 1.00 out of 1.00

What will be output for the following code?

```
if 3 == 3:
    print("Python is easy!")
```

a. Error

b. Python is easy! ✓

c. NO OUTPUT

Question **11**

Correct

Mark 1.00 out of 1.00

Can we write if/else into one line in python?

a. No

b. Yes ✓

Question **12**

Correct

Mark 1.00 out of 1.00

What is the output of the following code.

```
a=90
if a>100:
    if(a<=90 and a==90):
        print("REC")
    else:
        print("OPEN-ELECTIVE")
```

a. REC

b. REC
OPEN-ELECTIVE

c. No output ✓

d. OPEN-ELECTIVE

Question **13**

Correct

Mark 1.00 out of 1.00

What will be the output for the following code?

```
if  1-1:
   print("python")
else:
   print("0 is false")
```

a. python

b. 0 is false ✓

c. Error

Question **14**

Correct

Mark 1.00 out of 1.00

What keyword would you use to add an alternative condition to an if statement?

a. elseif

b. else if

c. elif ✓

Question **15**

Correct

Mark 1.00 out of 1.00

What will be the output for the following code?

```
if False:
    print("1001")
else:
    print("2002")
```

a. 2002 ✓

b. syntax error

c. 1001

Status	Finished
Started	Tuesday, 25 August 2026, 3:00 PM
Completed	Saturday, 29 August 2026, 4:14 PM
Duration	4 days 1 hour
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

A Number is said to be Disarium number when the sum of its digit raised to the power of their respective positions becomes equal to the number itself. Write a program to print number is Disarium or not.

Input Format:

Single Integer Input from stdin.

Output Format:

Yes or No.

Example Input:

175

Output:

Yes

Explanation

1^1 + 7^2 +5^3 = 175

Example Input:

123

Output:

No

For example:

Input	Result
175	Yes
123	No

Answer: (penalty regime: 0 %)

```
n=int(input())
temp=n
digits=len(str(n))
total=0
position=1
while temp>0:
    digit=temp%10
    total+=digit**digits
    digits-=1
    temp//=10
if total==n:
    print("Yes")
else:
    print("No")

```

	Input	Expected	Got	
✓	175	Yes	Yes	✓
✓	123	No	No	✓
✓	89	Yes	Yes	✓
✓	90	No	No	✓
✓	518	Yes	Yes	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

You are choreographing a circus show with various animals. For one act, you are given two kangaroos on a number line ready to jump in the positive direction.
•The first kangaroo starts at position x1 and moves at a speed v1 meters per jump.
•The second kangaroo starts at position x2 and moves at a speed of v2 meters per jump and x2 > x1
•You have to figure out to get both kangaroos at the same position at the same time as part of the show before k jumps. If it is possible, return YES, otherwise return NO.

Input Format:
x1-position of kangaroo1
v1-Speed of kangaroo1
x2-position of kangaroo2
v2-Speed of kangaroo2
k-jumps

Output Format:
Both kangaroos are at the same position within k jumps, YES, otherwise NO.

For example:

Input	Result
0 3 4 2 6	YES

Answer: (penalty regime: 0 %)

```
x1=int(input())
v1=int(input())
x2=int(input())
v2=int(input())
k=int(input())
if v1<=v2:
    print("NO")
else:
    distance=x2-x1
    speed_difference=v1-v2
    if distance%speed_difference==0:
        jumps=distance//speed_difference
        if jumps<k:
            print("YES")
        else:
            print("NO")
    else:
        print("NO")
```

	Input	Expected	Got	
✓	0 3 4 2 6	YES	YES	✓
✓	0 3 2 4 8	NO	NO	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00

Given a number N, find the next perfect square greater than N.

Input Format:

Integer input from stdin.

Output Format:

Perfect square greater than N.

Example Input:

10

Output:

16

For example:

Input	Result
10	16
20	25

Answer: (penalty regime: 0 %)

```
n=int(input())
i=1
while(i*i<=n):
    i+=1
print((i)*(i))
```

	Input	Expected	Got	
✓	10	16	16	✓
✓	20	25	25	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **4**

Correct

Mark 1.00 out of 1.00

Write a program that given an integer 'n', prints the number of integers that are less than or equal to 'n' and co-prime to 'n'

Two integers a and b are said to be relatively prime or co-prime if the only positive integer that evenly divides both of them is 1. That is, the only common positive factor of the two numbers is 1. This is equivalent to their greatest common divisor being 1.

Input Format:

One line containing the value of 'n', where 1<=n<=10,000

Output Format:

One line containing the number of integers that are co-prime to n and less than or equal to 'n'

Sample Test Cases

Test Case 1

Input

10

Output

4

Test Case 2

Input

23

Output

22

Test Case 3

Input

11

Output

10

Answer: (penalty regime: 0 %)

```
import math
n=int(input())
count=0
for i in range(1,n+1):
    if math.gcd(n,i)==1:
        count+=1
print(count)
```

	Input	Expected	Got	
✓	10	4	4	✓
✓	23	22	22	✓
✓	11	10	10	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **5**

Correct

Mark 1.00 out of 1.00

An automorphic number is a number whose square ends with the number itself.

For example, 5 is an automorphic number because 5*5 =25. The last digit is 5 which same as

the given number.

If it is an automorphic number display “Automorphic” else display “Not Automorphic”.

Input Format:

Take a Integer from Keyboard

Output Format:

Print Automorphic if given number is Automorphic number, otherwise Not Automorphic

Example input:

5

Output:

Automorphic

Example input:

25

Output:

Automorphic

Example input:

7

Output:

Not Automorphic

Answer: (penalty regime: 0 %)

```
a=int(input())
s=a*a
a=str(a)
s=str(s)
if s.endswith(a):
    print("Automorphic")
else:
    print("Not Automorphic")
```

	Input	Expected	Got	
✓	5	Automorphic	Automorphic	✓

	Input	Expected	Got	
✓	625	Automorphic	Automorphic	✓
✓	7	Not Automorphic	Not Automorphic	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **6**

Correct

Mark 1.00 out of 1.00

Write a program to find the count of the number of prime numbers in a specified range.

The starting and ending number of the range will be provided as input to the program.

Assumption: 2 <=starting number of the range<= ending number of the range<=7919

Example1: If the starting and ending number or the range is given as 2 and 20, the program must return 8, because there are 8 prime numbers in the specified range from 2 to 20. namely (2. 3. 5, 7, 11, 13, 17, 19)

Example2: If the starting and ending number of the range is given as 700 and 725, the program must return 3, because there are 3 prime numbers in the specified range from 700 to 725, namely (701, 709, 719)

For example:

Input	Result
2 20	8
700 725	3

Answer: (penalty regime: 0 %)

```
start=int(input())
end=int(input())
count=0
for num in range(start,end+1):
    if num<2:
        continue
    is_prime=True
    for i in range(2,int(num**0.5)+1):
        if num%i==0:
            is_prime=False
            break
    if is_prime:
        count+=1
print(count)
```

	Input	Expected	Got	
✓	2 20	8	8	✓
✓	700 725	3	3	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **7**

Correct

Mark 1.00 out of 1.00

Given a positive integer a, find the smallest positive integer b whose multiplication of each digit equals to a.

If there is no answer or the answer is not fit in 32-bit signed integer, then return 0.

Example 1

Input:

48

Output:

68

Example 2

Input:

15

Output:

35

For example:

Input	Result
48	68
15	35

Answer: (penalty regime: 0 %)

```
n=int(input())
if n<10:
    print(n)
else:
    ans=[]
for i in range(9,1,-1):
    while n%i==0:
        ans.append(str(i))
        n=n//i
if n!=1:
    print(0)
else:
    ans.reverse()
    for x in ans:
        print(x,end="")
```

```
15          print(x,end=" ")
```

	Input	Expected	Got	
✓	48	68	68	✓
✓	15	35	35	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **8**

Correct

Mark 1.00 out of 1.00

Write python program to print the following pattern based on input size.

Input:

3

Output:

```
  1
 2 3
4 5 6
```

For example:

Input	Result
4	1 2 3 4 5 6 7 8 9 10

Answer: (penalty regime: 0 %)

```
a=int(input())
num=1
for i in range(1,a+1):
    print(" "*(a-i),end="")
    for j in range(i):
        print(num,end=" ")
        num+=1
    print()
```

	Input	Expected	Got	
✓	3	1 2 3 4 5 6	1 2 3 4 5 6	✓
✓	4	1 2 3 4 5 6 7 8 9 10	1 2 3 4 5 6 7 8 9 10	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **9**

Correct

Mark 1.00 out of 1.00

Write a program to find the count of non-repeated digits in a given number N. The number will be passed to the program as an input of type int.

Assumption: The input number will be a positive integer number >= 1 and <= 25000.

Some examples are as below.

If the given number is 292, the program should return 1 because there is only 1 non-repeated digit '9' in this number

If the given number is 1015, the program should return 2 because there are 2 non-repeated digits in this number, '0', and '5'.

If the given number is 108, the program should return 3 because there are 3 non-repeated digits in this number, '1', '0', and '8'.

If the given number is 22, the function should return 0 because there are NO non-repeated digits in this number.

For example:

Input	Result
292	1
1015	2
108	3
22	0

Answer: (penalty regime: 0 %)

```
a=int(input())
count=[0]*10
for digit in str(a):
    count[int(digit)]+=1
ans=0
for i in range(10):
    if count[i]==1:
        ans+=1
print(ans)
```

	Input	Expected	Got	
✓	292	1	1	✓
✓	1015	2	2	✓
✓	108	3	3	✓
✓	22	0	0	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **10**

Correct

Mark 1.00 out of 1.00

Let's print a chessboard!

Write a program that takes input:

Integer N(represents the rows and columns of a chessboard) and also the starting character of the chessboard

Output Format

Print the chessboard as per the given examples

Sample Input / Output

Input:

2

W

Output:

WB

BW

Answer: (penalty regime: 0 %)

```
n=int(input())
c=str(input())
for i in range(n):
    for j in range(n):
        if(i+j)%2==0:
            print(c,end="")
        elif c=="W":
            print("B",end="")
        else:
            print("W",end="")
    print()
```

	Input	Expected	Got	
✓	2 W	WB BW	WB BW	✓
✓	3 B	BWB WBW BWB	BWB WBW BWB	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Status	Finished
Started	Tuesday, 15 September 2026, 3:46 PM
Completed	Tuesday, 15 September 2026, 4:04 PM
Duration	18 mins 11 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the output of the following code?

```
my_string = 'vijay'
for i in range(len(my_string)):
    print (my_string)
    my_string = 'a'
```

a. None

b. Error

c. vijay a a a a ✓

d. vaaaaaaaaaaa

Question **2**

Correct

Mark 1.00 out of 1.00

What is the output of the following Code?

str1="arvijayakumar"

print(str1[::-1])

Answer: ramukayajivra ✓

Question **3**

Correct

Mark 1.00 out of 1.00

What is the output of the following Code?

```
print(chr(70))
```

Answer: F ✓

Question **4**

Correct

Mark 1.00 out of 1.00

What is the output of the following Code?

```
str1="vijay"
print(str1.capitalize())
```

Answer: Vijay ✓

Question **5**

Correct

Mark 1.00 out of 1.00

What will be the output of below Python code?

```
str1="Application"
str2=str1.replace('a','A')
print(str2)
```

Answer: ApplicAtion ✓

Question **6**

Correct

Mark 1.00 out of 1.00

What is the output of the following Code?

```
str1="123456789"
print(str1[2:6:2])
```

Answer: 35 ✓

Question **7**

Correct

Mark 1.00 out of 1.00

What will be the output of below Python code?

```
str1="Information"
print(str1[2:8])
```

Answer: format ✓

Question **8**

Correct

Mark 1.00 out of 1.00

What is the output of the following code.

```
Line1 = "And Then There Were None"
Line2 = "Famous In Love"
Line3 = "Famous Were The Kol And Klaus"
Line4 = Line1 + Line2 + Line3
print("And" in Line4)
```

a. False

b. False 2

c. True 2

d. True ✓

Question **9**

Correct

Mark 1.00 out of 1.00

What will be the output of the following code?

```
a = '2'
b = '1'
print(a+b)
```

Answer: 21 ✓

Question **10**

Correct

Mark 1.00 out of 1.00

```
my_string = "arvjayakumar"
i = "i"
while i in my_string:
    print(i, end =" ")
```

a. i i i i i i . . .

b. a r v j a y a k u m a r

c. None ✓

d. arvjayakumar

Question **11**

Correct

Mark 1.00 out of 1.00

What is the output of the following code?

```
line = "I'll come by then."
eline = ""
for i in line:
    eline += chr(ord(i)+3)
print(eline)
```

a. L*oo frph e| wkhq1

b. L*oo#frph#e|#wkhq1 ✓

c. O*oo#Frph#E|#wKhq1

d. l*oo@frph@e|$wkhq1

Question **12**

Correct

Mark 1.00 out of 1.00

What is the output of the following code ?

```
print('raining'. find('z'))
```

a. Not Found

b. Type error

c. -1 ✓

d. ''

Question **13**

Correct

Mark 1.00 out of 1.00

What is the output of the following code?

```
print('*', "abcde".center(7), '*', sep='')
```

a. * abcde * ✓

b. *abcde *

c. * abcde*

d. *abcde*

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following is False?

a. String is immutable.

b. None of the mentioned

c. lower() function in string is used to return a string by converting the whole given string into lowercase.

d. capitalize() function in string is used to return a string by converting the whole given string into uppercase. ✓

Question **15**

Correct

Mark 1.00 out of 1.00

What arithmetic operators cannot be used with strings in Python?

a. +

b. All of the mentioned

c. - ✓

d. *

Status	Finished
Started	Tuesday, 22 September 2026, 3:58 PM
Completed	Wednesday, 23 September 2026, 10:34 PM
Duration	1 day 6 hours
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Given a string S which is of the format USERNAME@DOMAIN.EXTENSION, the program must print the EXTENSION, DOMAIN, USERNAME in the reverse order.

Input Format:

The first line contains S.

Output Format:

The first line contains EXTENSION.
The second line contains DOMAIN.
The third line contains USERNAME.

Boundary Condition:

1 <= Length of S <= 100

Example Input/Output 1:

Input:

abcd@gmail.com

Output:

com
gmail
abcd

For example:

Input	Result
arvijayakumar@rajalakshmi.edu.in	edu.in rajalakshmi arvijayakumar

Answer: (penalty regime: 0 %)

```
email=input()
username,domain_part=email.split("@")
domain,extension=domain_part.split(".",1)
print(extension)
print(domain)
print(username)
```

	Input	Expected	Got	
✓	abcd@gmail.com	com gmail abcd	com gmail abcd	✓
✓	arvijayakumar@rajalakshmi.edu.in	edu.in rajalakshmi arvijayakumar	edu.in rajalakshmi arvijayakumar	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

Assume that the given string has enough memory.

Don't use any extra space(IN-PLACE)

Sample Input 1

a2b4c6

Sample Output 1

aabbbbccccccc

Answer: (penalty regime: 0 %)

```
s=input()
i=0
ans=""
while i<len(s):
    ch=s[i]
    i+=1
    num=0
    while i<len(s) and s[i].isdigit():
        num=num*10+int(s[i])
        i+=1
    ans+=ch*num
print(ans)
```

	Input	Expected	Got	
✓	a2b4c6	aabbbbccccccc	aabbbbccccccc	✓
✓	a12b3d4	aaaaaaaaaaaabbbdddd	aaaaaaaaaaaabbbdddd	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00

Given a string s containing just the characters '(', ')', '{', '}', '[' and ']', determine if the input string is valid.

An input string is valid if:

Open brackets must be closed by the same type of brackets.

Open brackets must be closed in the correct order.

Constraints:

1 <= s.length <= 10^4

s consists of parentheses only '()[]{}'.

For example:

Test	Result
`print(ValidParenthesis("()"))`	`true`
`print(ValidParenthesis("()[]{}"))`	`true`
`print(ValidParenthesis("(]"))`	`false`

Answer: (penalty regime: 0 %)

Reset answer

```
def ValidParenthesis(s):
    stack=[]
    for ch in s:
        if ch=='(' or ch=='{' or ch=='[':
            stack.append(ch)
        else:
            if len(stack)==0:
                return "false"
            top=stack.pop()
            if ch==')' and top !='(':
                return "false"
            if ch=='}' and top !='{':
                return "false"
            if ch==']' and top !='[':
                return "false"
    if len(stack)==0:
        return "true"
    else:
        return "false"
```

	Test	Expected	Got	
✓	`print(ValidParenthesis("()"))`	`true`	`true`	✓
✓	`print(ValidParenthesis("()[]{}"))`	`true`	`true`	✓
✓	`print(ValidParenthesis("(]"))`	`false`	`false`	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **4**

Correct

Mark 1.00 out of 1.00

Given a string, determine if it is a palindrome, considering only alphanumeric characters and ignoring cases.

Note: For the purpose of this problem, we define empty string as valid palindrome.

Example 1:

```
Input:
A man, a plan, a canal: Panama

Output:
1
```

Example 2:

```
Input:
race a car

Output:
0
```

Constraints:

- `s` consists only of printable ASCII characters.

Answer: (penalty regime: 0 %)

```
s=input()
s=''.join(c.lower() for c in s if c.isalnum())
if s==s[::-1]:
    print(1)
else:
    print(0)
```

	Input	Expected	Got	
✓	A man, a plan, a canal: Panama	1	1	✓
✓	race a car	0	0	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **5**

Correct

Mark 1.00 out of 1.00

Find if a String2 is substring of String1. If it is, return the index of the first occurrence. else return -1.

Sample Input 1

thistest123string

123

Sample Output 1

8

Answer: (penalty regime: 0 %)

```
1 s1=input()
2 s2=input()
3 print(s1.find(s2))
```

	Input	Expected	Got	
✓	thistest123string 123	8	8	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **6**

Correct

Mark 1.00 out of 1.00

The program must accept **N** series of keystrokes as string values as the input. The character **^** represents undo action to clear the last entered keystroke. The program must print the string typed after applying the undo operations as the output. If there are no characters in the string then print **-1** as the output.

Boundary Condition(s):

1 <= N <= 100
1 <= Length of each string <= 100

Input Format:

The first line contains the integer N.
The next N lines contain a string on each line.

Output Format:

The first N lines contain the string after applying the undo operations.

Example Input/Output 1:

Input:

3
Hey ^ goooo^^glee^
lucke^y ^charr^ms
ora^^nge^^^^

Output:

Hey google
luckycharms
-1

Answer: (penalty regime: 0 %)

```
n=int(input())
for i in range(n):
    s=input()
    result = ""
    for ch in s:
            if ch == '^':
                result=result[:-1]
            else:
                result+=ch
    if result == "":
        print(-1)
    else:
        print(result)
```

	Input	Expected	Got	
✓	3 Hey ^ goooo^^glee^ lucke^y ^charr^ms ora^^nge^^^^	Hey google luckycharms -1	Hey google luckycharms -1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **7**

Correct

Mark 1.00 out of 1.00

Write a Python program to get one string and reverses a string. The input string is given as an array of characters `char[]` .

You may assume all the characters consist of printable ascii characters.

Example 1:

```
Input:
hello
Output:
olleh
```

Example 2:

```
Input:
Hannah
Output:
hannaH
```

Answer: (penalty regime: 0 %)

```
1 s=input()
2 r=s[::-1]
3 print(r)
```

	Input	Expected	Got	
✓	hello	olleh	olleh	✓
✓	Hannah	hannaH	hannaH	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **8**

Correct

Mark 1.00 out of 1.00

Consider the below words as key words and check the given input is key word or not.

keywords: {break, case, continue, default, defer, else, for, func, goto, if, map, range, return, struct, type, var}

Input format:

Take string as an input from stdin.

Output format:

Print the word is key word or not.

Example Input:

break

Output:

break is a keyword

Example Input:

IF

Output:

IF is not a keyword

For example:

Input	Result
break	break is a keyword
IF	IF is not a keyword

Answer: (penalty regime: 0 %)

```
keywords={"break","case","continue","default","defer","else","for","func","goto","if","map","range",
word=input()
if word in keywords:
    print(word,"is a keyword")
else:
    print(word,"is not a keyword")
```

	Input	Expected	Got	
✓	break	break is a keyword	break is a keyword	✓
✓	IF	IF is not a keyword	IF is not a keyword	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **9**

Correct

Mark 1.00 out of 1.00

Given a **non-empty** string s and an abbreviation abbr, return whether the string matches with the given abbreviation.

A string such as "word" contains only the following valid abbreviations:

["word", "1ord", "w1rd", "wo1d", "wor1", "2rd", "w2d", "wo2", "1o1d", "1or1", "w1r1", "1o2", "2r1", "3d", "w3", "4"]

Notice that only the above abbreviations are valid abbreviations of the string "word". Any other string is not a valid abbreviation of "word".

Note:

Assume s contains only lowercase letters and abbr contains only lowercase letters and digits.

Example 1:

Input

internationalization
i12iz4n

Output

true

Explanation

Given **s** = "internationalization", **abbr** = "i12iz4n":

Return true.

Example 2:

Input

apple
a2e

Output

false

Explanation

Given **s** = "apple", **abbr** = "a2e":

Return false.

Answer: (penalty regime: 0 %)

```
import sys
def solve():
    lines=sys.stdin.read().split()
    if len(lines) < 2:
        return
    s=lines[0]
    abbr=lines[1]
    i=0
    j=0
    n=len(s)
    m=len(abbr)
    while i < n and j < m:
        if abbr[j].isdigit():
            if abbr[j]=="0":
                print("false")
                return
            num=0
            while j < m and abbr[j].isdigit():
                num = num * 10 + int(abbr[j])
                j += 1
            i += num
        else:
            if s[i] != abbr[j]:
                print("false")
                return
            i += 1
            j += 1
    if i==n and j==m:
        print("true")
    else:
        print("false")
solve()
```

	Input	Expected	Got	
✓	internationalization i12iz4n	true	true	✓
✓	apple a2e	false	false	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **10**

Correct

Mark 1.00 out of 1.00

A pangram is a sentence where every letter of the English alphabet appears at least once.

Given a string sentence containing only lowercase English letters, return true if sentence is a pangram, or false otherwise.

Example 1:

Input:

thequickbrownfoxjumpsoverthelazydog

Output:

true

Explanation: sentence contains at least one of every letter of the English alphabet.

Example 2:

Input:

arvijayakumar

Output: false

Constraints:

1 <= sentence.length <= 1000

sentence consists of lowercase English letters.

For example:

Test	Result
`print(checkPangram('thequickbrownfoxjumpsoverthelazydog'))`	`true`
`print(checkPangram('arvijayakumar'))`	`false`

Answer: (penalty regime: 0 %)

Reset answer

```
def checkPangram(s):
    if len(set(s))==26:
        return "true"
    else:
        return "false"
```

	Test	Expected	Got	
✓	`print(checkPangram('thequickbrownfoxjumpsoverthelazydog'))`	`true`	`true`	✓
✓	`print(checkPangram('arvijayakumar'))`	`false`	`false`	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Status	Finished
Started	Thursday, 24 September 2026, 10:41 PM
Completed	Thursday, 24 September 2026, 10:50 PM
Duration	8 mins 29 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following function headers is correct?

a. def fun(a, b = 2, c = 3) ✓

b. def fun(a = 2, b = 3, c)

c. def fun(a, b, c = 3, d)

d. def fun(a = 2, b, c = 3)

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following number can never be generated by the following code: random.randrange(0, 100)

a. 99

b. 1

c. 100 ✓

d. 0

Question **3**

Correct

Mark 1.00 out of 1.00

Which keyword is used for defining a function?

a. Define

b. Function

c. def ✓

d. Fun

Question **4**

Correct

Mark 1.00 out of 1.00

Choose the incorrect statement.

a. print(pow(2, 3, 2))
b. print(pow(2.3, 3.2))
c. print(pow(2, 3))
d. None of the mentioned ✓

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following are advantages of using function in program?

a. It increases reusability.
b. It increases readability of program.
c. All of the mentioned ✓
d. It makes debugging easier.

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following statement is a function call?

a. sum ✓
b. function sum
c. call sum
d. def sum

Question **7**

Correct

Mark 1.00 out of 1.00

Write the output of : print(abs(-45))

a. 45.0
b. -45
c. 45 ✓
d. None of the mentioned

Question **8**

Correct

Mark 1.00 out of 1.00

def cal(n1) : What is n1?

a. Argument
b. None of the mentioned
c. Parameter ✓
d. Keyword

Question **9**

Correct

Mark 1.00 out of 1.00

Function defined to achieve some task as per the programmer's requirement is called a ____.

a. built in functions
b. All of the mentioned
c. user defined function ✓
d. library function

Question **10**

Correct

Mark 1.00 out of 1.00

6. Which of the following is not the built-in function?

a. print( )
b. input( )
c. tuple( )
d. dictionary( ) ✓

Question **11**

Correct

Mark 1.00 out of 1.00

What will be the output of the following Python code?

```
def maximum(x, y):
    if x > y:
        return x
    elif x == y:
        return 'The numbers are equal'
    else:
        return y
print(maximum(2, 3))
```

a. 2

b. 3 ✓

c. The numbers are equal

d. None of the mentioned

Question **12**

Correct

Mark 1.00 out of 1.00

Python function always returns a value

Select one:

True ✓

False

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following statement is not true regarding functions?

a. A function may or may not have parameters.

b. A function may or may not return value.

c. A function definition begins with "define" ✓

d. Function header always ends with a colon ( ) .

Question **14**

Correct

Mark 1.00 out of 1.00

Fill in the line of the following Python code for calculating the factorial of a number?

```
def factorial🦴 :
    if (n==1 or n==0):
       return 1
    else:
        return ——
num = 5;
print("number : ",num)
print("Factorial : ",factorial(num))
```

a. fact(n)*fact(n-1)

b. (n-1)*(n-2)

c. n*(n-1)

d. (n * factorial(n - 1)) ✓

Question **15**

Correct

Mark 1.00 out of 1.00

Write the output of : print(min(tuple("computer")))

a. o

b. c ✓

c. t

d. u

Status	Finished
Started	Thursday, 24 September 2026, 9:49 PM
Completed	Thursday, 24 September 2026, 10:11 PM
Duration	22 mins 33 secs
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 2.00 out of 2.00

Write a function that returns the value of a+aa+aaa+aaaa with a given digit as the value of a.

Suppose the following input is supplied to the program:

9

Then, the output should be:

9+99+999+9999=11106

Sample Input Format:

9

Sample Output format:

11106

For example:

Test	Result
print(Summation(8))	9872

Answer: (penalty regime: 0 %)

Reset answer

```
def Summation(n):
    total=0
    num=str(n)
    for i in range(1,5):
        total+=int(num*i)
    return total
```

	Test	Expected	Got	
✓	print(Summation(8))	9872	9872	✓
✓	print(Summation(10))	10203040	10203040	✓

Passed all tests! ✓

Correct

Marks for this submission: 2.00/2.00.

Question **2**

Correct

Mark 2.00 out of 2.00

complete function to implement coin change making problem i.e. finding the minimum

number of coins of certain denominations that add up to given amount of money.

The only available coins are of values 1, 2, 3, 4

Input Format:

Integer input from stdin.

Output Format:

return the minimum number of coins required to meet the given target.

Example Input:

16

Output:

4

Explanation:

We need only 4 coins of value 4 each

Example Input:

25

Output:

7

Explanation:

We need 6 coins of 4 value, and 1 coin of 1 value

Answer: (penalty regime: 0 %)

Reset answer

```
def coinChange(n):
    count=0
    count+=n//4
    n=n % 4
    count+=n//3
    n=n%3
    count+=n//2
    n=n%2
    count+=n
    return count
```

	Test	Expected	Got	
✓	print(coinChange(16))	4	4	✓

Passed all tests! ✓

Correct

Marks for this submission: 2.00/2.00.

Question **3**

Correct

Mark 2.00 out of 2.00

An e-commerce company plans to give their customers a special discount for Christmas.

They are planning to offer a flat discount. The discount value is calculated as the sum of all the prime digits in the total bill amount.

Write an algorithm to find the discount value for the given total bill amount.

Constraints

1 <= orderValue< 10e100000

Input

The input consists of an integer orderValue, representing the total bill amount.

Output

Print an integer representing the discount value for the given total bill amount.

Example Input

578

Output

12

For example:

Test	Result
`print(christmasDiscount(578))`	12

Answer: (penalty regime: 0 %)

Reset answer

```
def christmasDiscount(n):
    total=0
    for digit in str(n):
        if digit in "2357":
            total += int(digit)
    return total
```

	Test	Expected	Got	
✓	`print(christmasDiscount(578))`	12	12	✓

	Test	Expected	Got	
✓	`print(christmasDiscount(57))`	12	12	✓
✓	`print(christmasDiscount(222))`	6	6	✓
✓	`print(christmasDiscount(77777))`	35	35	✓

Passed all tests! ✓

Correct

Marks for this submission: 2.00/2.00.

Question **4**

Correct

Mark 2.00 out of 2.00

A number is considered to be ugly if its only prime factors are 2, 3 or 5.

[1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 15, ...] is the sequence of ugly numbers.

Task:

complete the function which takes a number n as input and checks if it's an ugly number.

return ugly if it is ugly, else return not ugly

Hint:

An ugly number U can be expressed as: U = 2^a * 3^b * 5^c, where a, b and c are nonnegative integers.

For example:

Test	Result
`print(checkUgly(6))`	`ugly`
`print(checkUgly(21))`	`not ugly`

Answer: (penalty regime: 0 %)

Reset answer

```
def checkUgly(n):
    if n <=0:
        return "not ugly"
    for p in [2,3,5]:
        while n%p==0:
            n=n//p
    if n==1:
        return "ugly"
    else:
        return "not ugly"
```

	Test	Expected	Got	
✓	`print(checkUgly(6))`	`ugly`	`ugly`	✓
✓	`print(checkUgly(21))`	`not ugly`	`not ugly`	✓

Passed all tests! ✓

Correct

Marks for this submission: 2.00/2.00.

Question **5**

Correct

Mark 2.00 out of 2.00

A strobogrammatic number is a number that looks the same when rotated 180 degrees (looked at upside down).

Write a program to determine if a number is strobogrammatic. The number is represented as a string.

Example 1:

Input:

69

Output:

true

Example 2:

Input:

88

Output:

true

Example 3:

Input:

962

Output:

false

Example 4:

Input:

1

Output:

true

For example:

Test	Result
`print(Strobogrammatic(69))`	`True`
`print(Strobogrammatic(962))`	`False`

Answer: (penalty regime: 0 %)

Reset answer

```
def Strobogrammatic(n):
    s=str(n)
    mp={'0':'0','1':'1','6':'9','8':'8','9':'6'}
    result=""
    for digit in reversed(s):
        if digit not in mp:
            return "False"
        result+=mp[digit]
    if result==s:
        return "True"
    else:
        return "False"
```

	Test	Expected	Got	
✓	`print(Strobogrammatic(69))`	`True`	`True`	✓
✓	`print(Strobogrammatic(88))`	`True`	`True`	✓
✓	`print(Strobogrammatic(962))`	`False`	`False`	✓

Passed all tests! ✓

Correct

Marks for this submission: 2.00/2.00.

Status	Finished
Started	Sunday, 4 October 2026, 5:59 PM
Completed	Sunday, 4 October 2026, 6:20 PM
Duration	21 mins 21 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What will be the output after the following statements?
m = [50, 25, 65, 0, 99]
n = max(m)
print👎

a. (50, 25, 65, 0, 99)

b. 99 ✓

c. 0

d. 25

Question **2**

Correct

Mark 1.00 out of 1.00

What will be the output after the following statements?
m = [5, 10, 35]
del m[:]
print(m)

a. [5, 35]

b. [5, 10, 35]

c. [] ✓

d. 5, 10, 35

Question **3**

Correct

Mark 1.00 out of 1.00

What will be the output after the following statements?
m = ['Games', 'in', 'Python']
n = 'Play' + m[0] + m[1] + m[2]
print👎

a. Games in Python

b. GamesinPython

c. Play Games in Python

d. PlayGamesinPython ✓

Question **4**

Correct

Mark 1.00 out of 1.00

Write the output of the following :

L = "123456"
L = list(L)
print(type(L[0]))

Answer: <class 'str'> ✓

Question **5**

Incorrect

Mark 0.00 out of 1.00

Find the output?

```
list3=[]
list1 ='REC_CSE_ECE'
list2= list1.split('_')
for i in list2:
    list3.extend(i)
print(list3)
```

a. Error

b. ['REC_CSE_ECE']

c. ['R', 'E', 'C', 'C', 'S', 'E', 'E', 'C', 'E']

d. ['REC', 'CSE', 'ECE'] ✗

Question **6**

Correct

Mark 1.00 out of 1.00

L=list("www.csiplearninghub.com")
print(L[20 : -1])

a. **['c' , 'o']** ✓

b. Error

c. **[c,o,m]**

Question **7**

Correct

Mark 1.00 out of 1.00

L=[0.5 * x for x in range(4)]

print(L)

Answer: [0.0, 0.5, 1.0, 1.5] ✓

Question **8**

Correct

Mark 1.00 out of 1.00

Suppose list1 is [3, 4, 5, 20, 5], what is list1.index(5)?

a. 4

b. -3 ✓

c. 1

Question **9**

Correct

Mark 1.00 out of 1.00

What will be the data type of m after the following statement?
m = [90, 'A', 115, 'B', 250]

a. Dictionary

b. Tuple

c. String

d. List ✓

Question **10**

Correct

Mark 1.00 out of 1.00

Fill in the blanks with same word in both places

```
>>> import _________
>>> list1 = [1,2,3,4,5]
>>> list2 = _________copy(list1)
>>> list2
```

a. Copy ✓

b. Math

c. Pickle

Question **11**

Correct

Mark 1.00 out of 1.00

Find the output?

```
list3=[]
list1 ='REC_CSE_ECE'
list2= list1.split('_')
for i in list2:
    list3.extend(i)
print(len(list3))
```

a. 9 ✓

b. 3

c. 11

d. 12

Question **12**

Correct

Mark 1.00 out of 1.00

Find the output?

```
list1 = [1, 2, 3, 4,1,2,3]
print(list1.pop())
```

a. 1

b. [ ]

c. 2

d. 3 ✓

Question **13**

Correct

Mark 1.00 out of 1.00

What will be the output after the following statements?
m = ['July', 'September', 'December']
n = m[0] + m[2]
print(n)

a. July

b. JulyDecember ✓

c. JulySeptember

d. SeptemberDecember

Question **14**

Correct

Mark 1.00 out of 1.00

What will be the output after the following statements?
m = [25, 34, 70, 63]
n = str(m[1]) +str(m[2])
print(n)

a. 95

b. 104

c. 3470 ✓

d. 2534

Question **15**

Correct

Mark 1.00 out of 1.00

What will be the output after the following statements?
m = list(range(7,10))
print(m)

a. list([7, 8, 9])

b. [7, 8, 9, 10]

c. [7, 8, 9] ✓

d. 789

Status	Finished
Started	Saturday, 10 October 2026, 1:41 PM
Completed	Saturday, 10 October 2026, 2:22 PM
Duration	40 mins 59 secs
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Given two arrays of positive integers, for each element in the second array, find the total number of elements in the first array which are *less than or equal to* that element. Store the values determined in an array.

For example, if the first array is *[1, 2, 3]* and the second array is *[2, 4]*, then there are *2* elements in the first array *less than or equal to 2*. There are *3* elements in the first array which are *less than or equal to 4*. We can store these answers in an array, *answer = [2, 3]*.

Program Description

The program must return an array of *m* positive integers, one *for each maxes[i]* representing the total number of elements *nums[j]* satisfying *nums[j] ≤ maxes[i]* where $0 \le j < n$ and $0 \le i < m$, in the given order.

The program has the following:

nums[nums[0],...nums[n-1]]: first array of positive integers
maxes[maxes[0],...maxes[n-1]]: second array of positive integers

Constraints

- $2 \le n, m \le 10^5$
- $1 \le nums[j] \le 10^9$, where $0 \le j < n$.
- $1 \le maxes[i] \le 10^9$, where $0 \le i < m$.

Input Format For Custom Testing

Input from stdin will be processed as follows and passed to the program.

The first line contains an integer *n*, the number of elements in *nums*.
The next *n* lines each contain an integer describing *nums[j]* where $0 \le j < n$.
The next line contains an integer *m*, the number of elements in *maxes*.
The next *m* lines each contain an integer describing *maxes[i]* where $0 \le i < m$.

Sample Case 0

Sample Input 0

```
4
1
4
2
4
2
3
5
```

Sample Output 0

```
2
4
```

Explanation 0

We are given *n = 4*, *nums = [1, 4, 2, 4]*, *m = 2*, and *maxes = [3, 5]*.

1. For *maxes[0] = 3*, we have *2* elements in *nums* (*nums[0] = 1* and *nums[2] = 2*) that are ≤ *maxes[0]*.
2. For *maxes[1] = 5*, we have *4* elements in *nums* (*nums[0] = 1*, *nums[1] = 4*, *nums[2] = 2*, and *nums[3] = 4*) that are ≤ *maxes[1]*.

Thus, the program returns the array *[2, 4]* as the answer.

Sample Case 1

Sample Input 1

```
5
2
10
5
4
8
4
3
1
7
8
```

Sample Output 1

```
1
0
3
4
```

Explanation 1

We are given, *n = 5*, *nums = [2, 10, 5, 4, 8]*, *m = 4*, and *maxes = [3, 1, 7, 8]*.

1. For *maxes[0] = 3*, we have *1* element in *nums* (*nums[0] = 2*) that is ≤ *maxes[0]*.
2. For *maxes[1] = 1*, there are *0* elements in *nums* that are ≤ *maxes[1]*.
3. For *maxes[2] = 7*, we have *3* elements in *nums* (*nums[0] = 2*, *nums[2] = 5*, and *nums[3] = 4*) that are ≤ *maxes[2]*.
4. For *maxes[3] = 8*, we have *4* elements in *nums* (*nums[0] = 2*, *nums[2] = 5*, *nums[3] = 4*, and *nums[4] = 8*) that are ≤ *maxes[3]*.

Thus, the program returns the array *[1, 0, 3, 4]* as the answer.

Answer: (penalty regime: 0 %)

```
n=int(input())
nums=[]
for i in range(n):
    nums.append(int(input()))
m=int(input())
maxes=[]
for i in range(m):
    maxes.append(int(input()))
ans=[]
for x in maxes:
    count=0
    for y in nums:
        if y <= x:
            count=count+1
    ans.append(count)
for value in ans:
    print(value)
```

	Input	Expected	Got	
✓	4 1 4 2 4 2 3 5	2 4	2 4	✓
✓	5 2 10 5 4 8 4 3 1 7 8	1 0 3 4	1 0 3 4	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

Complete the program to count frequency of each element of an array. Frequency of a particular element will be printed once.

Sample Test Cases

Test Case 1

Input

7
23
45
23
56
45
23
40

Output

23 occurs 3 times
45 occurs 2 times
56 occurs 1 times
40 occurs 1 times

Answer: (penalty regime: 0 %)

```
n=int(input())
arr=[]
for i in range(n):
    arr.append(int(input()))
for i in range(n):
    if arr[i] not in arr[:i]:
        count=arr.count(arr[i])
        print(arr[i],"occurs",count,"times")
```

	Input	Expected	Got	
✓	7 23 45 23 56 45 23 40	23 occurs 3 times 45 occurs 2 times 56 occurs 1 times 40 occurs 1 times	23 occurs 3 times 45 occurs 2 times 56 occurs 1 times 40 occurs 1 times	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00

Assume you have an array of length ***n*** initialized with all **0**'s and are given ***k*** update operations.

Each operation is represented as a triplet: **[startIndex, endIndex, inc]** which increments each element of subarray **A[startIndex ... endIndex]** (startIndex and endIndex inclusive) with **inc**.

Return the modified array after all ***k*** operations were executed.

Example:

Input:

5

3

1 3 2

2 4 3

0 2 -2

Output:

-2 0 3 5 3

Explanation:

Initial state:

length = 5, updates = [[1,3,2],[2,4,3],[0,2,-2]]

[0,0,0,0,0]

After applying operation [1,3,2]:

[0,2,2,2,0]

After applying operation [2,4,3]:

[0,2,5,5,3]

After applying operation [0,2,-2]:

[-2,0,3,5,3]

Answer: (penalty regime: 0 %)

```
n=int(input())
k=int(input())
arr=[0]*n
for i in range(k):
    start,end,inc=map(int,input().split())
    for i in range(start,end+1):
        arr[i]+=inc
print(*arr)
```

	Input	Expected	Got	
✓	5 3 1 3 2 2 4 3 0 2 -2	-2 0 3 5 3	-2 0 3 5 3	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **4**

Correct

Mark 1.00 out of 1.00

Determine the factors of a number (i.e., all positive integer values that evenly divide into a number) and then return the pth element of the list, sorted ascending. If there is no p^{th} element, return 0.

Example

n = 20

p = 3

The factors of 20 in ascending order are {1, 2, 4, 5, 10, 20}. Using 1-based indexing, if p = 3, then 4 is returned. If p > 6, 0 would be returned.

Constraints

$1 \leq n \leq 10^{15}$

$1 \leq p \leq 10^{9}$

The first line contains an integer n, the number to factor.

The second line contains an integer p, the 1-based index of the factor to return.

Sample Case 0

Sample Input 0

10

3

Sample Output 0

5

Explanation 0

Factoring n = 10 results in {1, 2, 5, 10}. Return the p = 3rd factor, 5, as the answer.

Sample Case 1

Sample Input 1

10

5

Sample Output 1

0

Explanation 1

Factoring n = 10 results in {1, 2, 5, 10}. There are only 4 factors and p = 5, therefore 0 is returned as the answer.

Sample Case 2

Sample Input 2

1

1

Sample Output 2

1

Explanation 2

Factoring n = 1 results in {1}. The p = 1st factor of 1 is returned as the answer.

For example:

Input	Result
10 3	5
10 5	0
1 1	1

Answer: (penalty regime: 0 %)

```
n=int(input())
p=int(input())
count=0
for i in range(1,n+1):
    if n%i==0:
        count+=1
        if count==p:
            print(i)
            break
else:
    print(0)
```

	Input	Expected	Got	
✓	10 3	5	5	✓
✓	10 5	0	0	✓
✓	1 1	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **5**

Correct

Mark 1.00 out of 1.00

An array is monotonic if it is either **monotone increasing** or **monotone decreasing**.
An array A is monotone increasing if for all i <= j, **A[i] <= A[j]**. An array A is monotone decreasing if for all i <= j, **A[i] >= A[j]**.

Write a program if n array is monotonic or not. Print "True" if is monotonic or "False" if it is not. Array can be monotone increasing or decreasing.

Input Format:

First line n-get number of elements

Next n Lines is the array of elements

Output Format:

True ,if array is monotone increasing or decreasing.

otherwise False is printed

Sample Input1

4

5

6

7

8

Sample Output1

True

Sample Input2

4

6

5

4

3

Sample Output2

True

Sample Input 3

4

6

7

8

7

Sample Output3

False

For example:

Input	Result
4 6 5 4 3	True

Answer: (penalty regime: 0 %)

```
n=int(input())
a=[]
for i in range(n):
    a.append(int(input()))
x = a.copy()
x.sort()
y=a.copy()
y.sort(reverse=True)
if a==x or a==y:
    print("True")
else:
    print("False")
```

	Input	Expected	Got	
✓	4 6 5 4 3	True	True	✓
✓	4 3 5 7 4	False	False	✓
✓	4 1 6 9 2	False	False	✓
✓	4 9 6 4 2	True	True	✓
✓	3 2 1 4	False	False	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **6**

Correct

Mark 1.00 out of 1.00

The program must accept **N** integers and an integer **K** as the input. The program must print every K integers in descending order as the output.

Note: If **N % K != 0**, then sort the final N%K integers in descending order.

Boundary Condition(s):

1 <= N <= 10^4
-99999 <= Array Element Value <= 99999

Input Format:

The first line contains the values of N and K separated by a space.
The second line contains N integers separated by space(s).

Output Format:

The first line contains N integers.

Example Input/Output 1:

Input:

7 3
48 541 23 68 13 41 6

Output:

541 48 23 68 41 13 6

Explanation:

The first three integers are 48 541 23, after sorting in descending order the integers are **541 48 23**.
The second three integers are 68 13 41, after sorting in descending order the integers are **68 41 13**.
The last integer is **6**.
The integers are **541 48 23 68 41 13 6**
Hence the output is **541 48 23 68 41 13 6**.

Answer: (penalty regime: 0 %)

```
n,k=map(int,input().split())
a=list(map(int,input().split()))
for i in range(0,n,k):
    group=a[i:i+k]
    group.sort(reverse=True)
    a[i:i+k]=group
print(*a)
```

```
print(a)
```

	Input	Expected	Got	
✓	7 3 48 541 23 68 13 41 6	541 48 23 68 41 13 6	541 48 23 68 41 13 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **7**

Correct

Mark 1.00 out of 1.00

Program to print all the distinct elements in an array. Distinct elements are nothing but the unique (non-duplicate) elements present in the given array.

Input Format:

First line take an Integer input from stdin which is array length n.

Second line take n Integers which is inputs of array.

Output Format:

Print the Distinct Elements in Array in single line which is space Separated

Example Input:

5

1

2

2

3

4

Output:

1 2 3 4

Example Input:

6

1

1

2

2

3

3

Output:

1 2 3

For example:

Input	Result
5 1 2 2 3 4	1 2 3 4
6 1 1 2 2 3 3	1 2 3

Answer: (penalty regime: 0 %)

```
n=int(input())
arr=[]
for i in range(n):
    arr.append(int(input()))
distinct=[]
for x in arr:
    if x not in distinct:
        distinct.append(x)
print(*distinct)
```

	Input	Expected	Got	
✓	5 1 2 2 3 4	1 2 3 4	1 2 3 4	✓
✓	6 1 1 2 2 3 3	1 2 3	1 2 3	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **8**

Correct

Mark 1.00 out of 1.00

Given a matrix mat where every row is sorted in **strictly increasing** order, return the **smallest common element** in all rows.

If there is no common element, return -1.

Example 1:

Input:

4 5

1 2 3 4 5

2 4 5 8 10

3 5 7 9 11

1 3 5 7 9

Output:

5

Constraints:

- 1 <= mat.length, mat[i].length <= 500
- 1 <= mat[i][j] <= 10^4
- mat[i] is sorted in strictly increasing order.

Answer: (penalty regime: 0 %)

```
r,c=map(int,input().split())
a=[list(map(int,input().split())) for i in range(r)]
ans=-1
for x in a[0]:
    if all(x in row for row in a):
        ans=x
        break
print(ans)
```

	Input	Expected	Got	
✓	4 5 1 2 3 4 5 2 4 5 8 10 3 5 7 9 11 1 3 5 7 9	5	5	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **9**

Correct

Mark 1.00 out of 1.00

Given an integer n, return an list of length n + 1 such that for each i (0 <= i <= n), ans[i] is the number of 1's in the binary representation of i.

Example:

```
Input: n = 2
Output: [0,1,1]
Explanation:
0 --> 0
1 --> 1
2 --> 10
```

Example2:

```
Input: n = 5
Output: [0,1,1,2,1,2]
Explanation:
0 --> 0
1 --> 1
2 --> 10
3 --> 11
4 --> 100
5 --> 101
```

Note: Complete the given function alone

For example:

Test	Result
print(CountingBits(5))	[0, 1, 1, 2, 1, 2]

Answer: (penalty regime: 0 %)

Reset answer

```
def CountingBits(n):
    ans=[]
    for i in range(n+1):
        ans.append(bin(i).count('1'))
    return ans
```

	Test	Expected	Got	
✓	`print(CountingBits(2))`	`[0, 1, 1]`	`[0, 1, 1]`	✓
✓	`print(CountingBits(5))`	`[0, 1, 1, 2, 1, 2]`	`[0, 1, 1, 2, 1, 2]`	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **10**

Correct

Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[i] - A[j] = k, i != j.

Input Format

1. First line is number of test cases T. Following T lines contain:
2. N, followed by N integers of the array
3. The non-negative integer k

Output format

Print 1 if such a pair exists and 0 if it doesn't.

Example

Input

1

3

1

3

5

4

Output:

1

Input

1

3

1

3

5

99

Output

0

For example:

Input	Result
1 3 1 3 5 4	1
1 3 1 3 5 99	0

Answer: (penalty regime: 0 %)

```
t=int(input())
for i in range(t):
    n=int(input())
    a=[]
    for i in range(n):
        a.append(int(input()))
    k=int(input())
    ans=0
    for i in range(n):
        for j in range(i+1,n):
            if a[j]-a[i]==k:
                ans=1
    print(ans)
```

	Input	Expected	Got	
✓	1 3 1 3 5 4	1	1	✓
✓	1 3 1 3 5 99	0	0	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Status	Finished
Started	Saturday, 10 October 2026, 2:31 PM
Completed	Saturday, 10 October 2026, 2:47 PM
Duration	15 mins 20 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Find the output of the given Python program?

```
t1 = (55, 44, 33, 22, 11)
x = [t1[i] for i in range(0, len(t1), 2)]
print(x)
```

a. ([55,33,11])

b. [(55,33,11)]

c. [55, 33, 11] ✓

d. (55,33,11)

Question **2**

Correct

Mark 1.00 out of 1.00

What will be the output of following Python code?

```
set1={2,5,3}

set2={3,1}

set3={}

set3=set1&set2

print(set3)
```

a. {2,5,1}

b. {}

c. {2,5,3,1}

d. {3} ✓

Question **3**

Correct

Mark 1.00 out of 1.00

If a=(15,16,17,18,19,25), then a[1:-1] will be

Note :

a=(15,16,17,18,19,25)

print((a[1:-1]))

a. (25,19,18,17)

b. (16,17,18)

c. (16,17,18,19) ✓

d. Error

Question **4**

Correct

Mark 1.00 out of 1.00

What is the output of the given below program?

t = (58, 47, 36, 25, 14, 3)

x = t[2:-1]

print(x)

a. (58,47,36,25)

b. (36, 25, 14) ✓

c. (3,14,25)

d. Error

Question **5**

Correct

Mark 1.00 out of 1.00

What will be printed when the following code executes?

```
a = ("Python Programming")
```

```
print type(a)
```

```

```

a. <class 'tuple'>

b. <class 'int'>

c. <class 'str'> ✓

d. str

Question **6**

Correct

Mark 1.00 out of 1.00

```
What is the output of the following code?
```

```
aTuple = (10, 20, 30, 40, 50, 60, 70, 80)
print(aTuple[2:5], aTuple[:4], aTuple[3:])
```

a. (30, 40, 50)(40, 50, 60, 70, 80)

b. (30, 40, 50) (10, 20, 30, 40) (40, 50, 60, 70, 80) ✓

c. (10, 20, 30, 40) (40, 50, 60, 70, 80)

d. (30, 40, 50) (10, 20, 30, 40)

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following options will not result in an error when performed on tuples in Python where tupl=(5,2,7,0,3)?

a. tupl[1]=2

b. tupl.sort()

c. tupl.append(2)

d. tupl1=tupl+tupl ✓

Question **8**

Correct

Mark 1.00 out of 1.00

Write the Output of the following Code?

t = (15,83,83,52,60,45,52,85,100)

print(min(t)+max(t)+t.count(52))

a. 2

b. 117 ✓

c. Error

d. 100

Question **9**

Correct

Mark 1.00 out of 1.00

What is printed when the following code is run?

```
tup = ('30', '3', '2', '8')
print(sorted(tup,reverse = True))
```

a. ['2', '3', '30', '8']

b. ['2', '3', '8', '30']

c. ['8', '30', '3', '2'] ✓

d. ['30', '8', '3', '2']

Question **10**

Correct

Mark 1.00 out of 1.00

What is the output of the given below program?

```
t1 = (1,2,3)
t2 = (4,5,6)
x = t1+t2
print(x)
```

a. (1,2,3,4,5,6) ✓

b. (1,2,3)(4,5,6)

c. Error

d. (1,2,3,3,2,1)

Question **11**

Correct

Mark 1.00 out of 1.00

What is the output of the following union operation

```
set1 = {10, 20, 30, 40}
set2 = {50, 20, "10", 60}

set3 = set1.union(set2)
print(set3)
```

a. {40, '10', 50, 20, 60, 30}

b. {40, 10, 50, 20, 60, 30}

c. {40, 10, '10', 50, 20, 60, 30} ✓

d. SynatxError: Different types cannot be used with sets

Question **12**

Correct

Mark 1.00 out of 1.00

Select all the correct options to remove "ECE" from the set.

```
sampleSet = {"ECE", "R&A", "MCT"}
```

a. remove.sampleSet("ECE")

b. del.sampleSet("ECE")

c. sampleSet.discard("ECE") ✓

d. sampleSet.delete("ECE")

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following options will produce the same output?

```
t = (15, 83, 21, 49, 60,45,52,85,100)
# options i, ii, iii, or iv
print(t[:-1])
print(t[0:5])
print(t[0:8])
print(t[-7:])
```

a. ii,iv

b. i,ii

c. i,iii ✓

d. iii,iv

Question **14**

Correct

Mark 1.00 out of 1.00

Choose the correct option.

a. In Python, a tuple can contain both integers and strings as its elements. ✓

b. In Python, a tuple can contain only integers as its elements.

c. In Python, a tuple can contain only strings as its elements.

d. In Python, a tuple can contain either string or integer but not both at a time.

Question **15**

Correct

Mark 1.00 out of 1.00

What is the output of the given below program?

```
my_t1 = (1, 2, 3, 4)
my_t1.append( (5, 6, 7) )
print(len(my_t1))
```

a. 5

b. 2

c. 1

d. Error ✓

Status	Finished
Started	Saturday, 10 October 2026, 3:01 PM
Completed	Saturday, 10 October 2026, 4:01 PM
Duration	1 hour
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

The **DNA sequence** is composed of a series of nucleotides abbreviated as `'A'`, `'C'`, `'G'`, and `'T'`.

- For example, `"ACGAATTCCG"` is a **DNA sequence**.

When studying **DNA**, it is useful to identify repeated sequences within the DNA.

Given a string `s` that represents a **DNA sequence**, return all the `10`**-letter-long** sequences (substrings) that occur more than once in a DNA molecule. You may return the answer in **any order**.

Example 1:

```
Input: s = "AAAAACCCCCAAAAACCCCCCAAAAAGGGTTT"
Output: ["AAAAACCCCC","CCCCCAAAAA"]
```

Example 2:

```
Input: s = "AAAAAAAAAAAAA"
Output: ["AAAAAAAAAA"]
```

For example:

Input	Result
AAAAACCCCCAAAAACCCCCCAAAAAGGGTTT	AAAAACCCCC CCCCCAAAAA

Answer: (penalty regime: 0 %)

```
s=input()
seen=set()
repeated=set()
for i in range(len(s)-9):
    seq=s[i:i+10]
    if seq in seen:
        repeated.add(seq)
    else:
        seen.add(seq)
for seq in sorted(repeated):
    print(seq)
```

	Input	Expected	Got	
✓	AAAAACCCCCAAAAACCCCCCAAAAAGGGTTT	AAAAACCCCC CCCCCAAAAA	AAAAACCCCC CCCCCAAAAA	✓

	Input	Expected	Got	
✓	AAAAAAAAAAAA	AAAAAAAAAA	AAAAAAAAAA	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

Check if a set is a subset of another set.

Example:

Sample Input1:

mango apple

mango orange

mango

output1:

yes

set3 is subset of set1 and set2

```
input2:
```

```
mango orange
```

```
banana orange
```

```
grapes
```

```
output2:
```

no

For example:

Test	Input	Result
1	mango apple mango orange mango	yes set3 is subset of set1 and set2
2	mango orange banana orange grapes	No

Answer: (penalty regime: 0 %)

```
def check_subset():
    set1=set(input().split())
    set2=set(input().split())
    set3=set(input().split())
    if set3.issubset(set1) and set3.issubset(set2):
        print("yes")
        print("set3 is subset of set1 and set2")
    else:
        print("No")
if __name__=="__main__":
    check_subset()
```

	Test	Input	Expected	Got	
✓	1	mango apple mango orange mango	yes set3 is subset of set1 and set2	yes set3 is subset of set1 and set2	✓
✓	2	mango orange banana orange grapes	No	No	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **3**

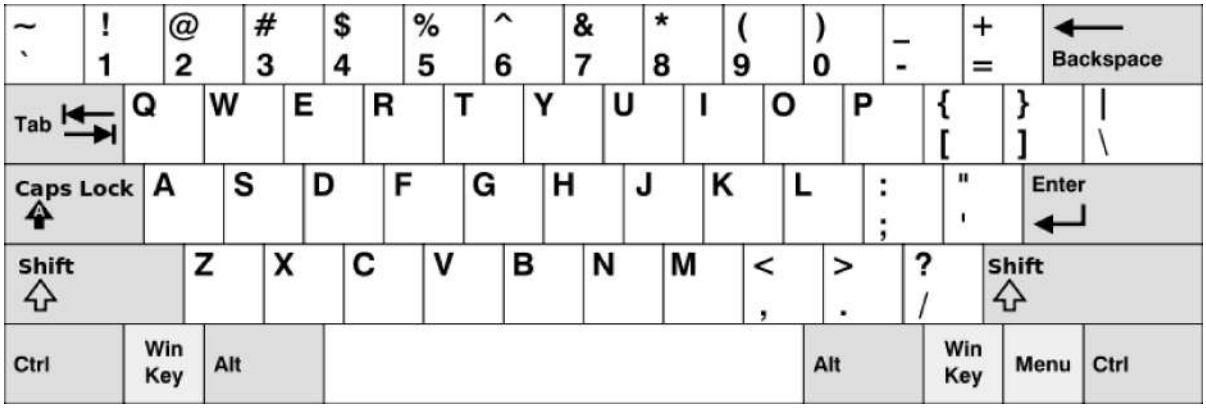
Correct

Mark 1.00 out of 1.00

Given an array of strings `words`, return *the words that can be typed using letters of the alphabet on only one row of American keyboard like the image below.*

In the **American keyboard**:

- the first row consists of the characters `"qwertyuiop"`,
- the second row consists of the characters `"asdfghjkl"`, and
- the third row consists of the characters `"zxcvbnm"`.



Example 1:

```
Input: words = ["Hello","Alaska","Dad","Peace"]
Output: ["Alaska","Dad"]
```

Example 2:

```
Input: words = ["omk"]
Output: []
```

Example 3:

```
Input: words = ["adsdf","sfd"]
Output: ["adsdf","sfd"]
```

For example:

Input	Result
4 Hello Alaska Dad Peace	Alaska Dad
2 adsfd afd	adsfd afd

Answer: (penalty regime: 0 %)

```
def findWords(words):
    rows=[set("qwertyuiop"),set("asdfghjkl"),set("zxcvbnm")]
    result=[]
    for word in words:
        w=word.lower()
```

```
        for row in rows:
            if all(ch in row for ch in w):
                result.append(word)
                break
    return result
n=int(input())
words=[]
for i in range(n):
    words.append(input())
result=findWords(words)
if result:
    print(*result,sep="\n")
else:
    print("No words")
```

	Input	Expected	Got	
✓	4 Hello Alaska Dad Peace	Alaska Dad	Alaska Dad	✓
✓	1 omk	No words	No words	✓
✓	2 adsfd afd	adsfd afd	adsfd afd	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **4**

Correct

Mark 1.00 out of 1.00

Given a tuple and a positive integer k, the task is to find the count of distinct pairs in the tuple whose sum is equal to **K**.

Examples:

> **Input:** t = (5, 6, 5, 7, 7, 8), K = 13
> **Output:** 2
> **Explanation:**
> Pairs with sum K(= 13) are {(5, 8), (6, 7), (6, 7)}.
> Therefore, distinct pairs with sum K(= 13) are { (5, 8), (6, 7) }.
> Therefore, the required output is 2.

For example:

Input	Result
1,2,1,2,5 3	1
1,2 0	0

Answer: (penalty regime: 0 %)

```
t=tuple(map(int,input().split(',')))
k=int(input())
pairs=set()
for i in range(len(t)):
    for j in range(i+1,len(t)):
        if t[i] + t[j] == k:
            pairs.add(tuple(sorted((t[i],t[j]))))
print(len(pairs))
```

	Input	Expected	Got	
✓	5,6,5,7,7,8 13	2	2	✓
✓	1,2,1,2,5 3	1	1	✓
✓	1,2 0	0	0	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **5**

Correct

Mark 1.00 out of 1.00

There is a malfunctioning keyboard where some letter keys do not work. All other keys on the keyboard work properly.

Given a string text of words separated by a single space (no leading or trailing spaces) and a string brokenLetters of all distinct letter keys that are broken, return the number of words in text you can fully type using this keyboard.

Example 1:

Input: text = "hello world", brokenLetters = "ad"

Output:

1

Explanation: We cannot type "world" because the 'd' key is broken.

For example:

Input	Result
hello world ad	1
Faculty Upskilling in Python Programming ak	2

Answer: (penalty regime: 0 %)

```
text=input()
brokenLetters=input().lower()
count=0
for word in text.split():
    can_type=True
    for letter in brokenLetters:
        if letter in word.lower():
            can_type=False
            break
    if can_type:
        count+=1
print(count)
```

	Input	Expected	Got	
✓	hello world ad	1	1	✓
✓	Welcome to REC e	1	1	✓
✓	Faculty Upskilling in Python Programming ak	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **6**

Correct

Mark 1.00 out of 1.00

Program to print all the distinct elements in an array. Distinct elements are nothing but the unique (non-duplicate) elements present in the given array.

Input Format:

First line take an Integer input from stdin which is array length n.

Second line take n Integers which is inputs of array.

Output Format:

Print the Distinct Elements in Array in single line which is space Separated

Example Input:

5

1 2 2 3 4

Output:

1 2 3 4

Example Input:

6

1 1 2 2 3 3

Output:

1 2 3

For example:

Input	Result
5 1 2 2 3 4	1 2 3 4

Answer: (penalty regime: 0 %)

```
n=int(input())
arr=[]
for i in range(n):
    arr.append(int(input()))
distinct=[]
for x in arr:
    if x not in distinct:
        distinct.append(x)
print(*distinct)
```

	Input	Expected	Got	
✓	5 1 2 2 3 4	1 2 3 4	1 2 3 4	✓
✓	6 1 1 2 2 3 3	1 2 3	1 2 3	✓
✓	5 11 22 11 22 11	11 22	11 22	✓
✓	10 1 2 3 4 5 1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **7**

Correct

Mark 1.00 out of 1.00

Write a program to eliminate the common elements in the given 2 arrays and print only the non-repeating elements and the total number of such non-repeating elements.

Input Format:

The first line contains space-separated values, denoting the size of the two arrays in integer format respectively.

The next two lines contain the space-separated integer arrays to be compared.

Sample Input:

5 4

1 2 8 6 5

2 6 8 10

Sample Output:

1 5 10

3

Sample Input:

5 5

1 2 3 4 5

1 2 3 4 5

Sample Output:

NO SUCH ELEMENTS

For example:

Input	Result
5 4 1 2 8 6 5 2 6 8 10	1 5 10 3
5 5 1 2 3 4 5 1 2 3 4 5	NO SUCH ELEMENTS

Answer: (penalty regime: 0 %)

```
n,m=map(int,input().split())
a=list(map(int,input().split()))
b=list(map(int,input().split()))
result=[]
for x in a:
    if x not in b:
        result.append(x)
for x in b:
    if x not in a:
        result.append(x)
if result:
    print(*result)
    print(len(result))
else:
    print("NO SUCH ELEMENTS")
```

	Input	Expected	Got	
✓	5 4 1 2 8 6 5 2 6 8 10	1 5 10 3	1 5 10 3	✓
✓	3 3 10 10 10 10 11 12	11 12 2	11 12 2	✓
✓	5 5 1 2 3 4 5 1 2 3 4 5	NO SUCH ELEMENTS	NO SUCH ELEMENTS	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **8**

Correct

Mark 1.00 out of 1.00

Coders here is a simple task for you, Given string str. Your task is to check whether it is a binary string or not by using python set.

Examples:

Input: str = "01010101010"

Output: Yes

Input: str = "REC101"

Output: No

For example:

Input	Result
01010101010	Yes
010101 10101	No

Answer: (penalty regime: 0 %)

```
s=input().strip()
if set(s).issubset({'0','1'}):
    print("Yes")
else:
    print("No")
```

	Input	Expected	Got	
✓	01010101010	Yes	Yes	✓
✓	REC123	No	No	✓
✓	010101 10101	No	No	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **9**

Correct

Mark 1.00 out of 1.00

Given an array of integers `nums` containing `n + 1` integers where each integer is in the range `[1, n]` inclusive.There is only **one repeated number** in `nums`, return *this repeated number*. Solve the problem using set.

Example 1:

```
Input: nums = [1,3,4,2,2]
```

```
Output: 2
```

Example 2:

```
Input: nums = [3,1,3,4,2]
```

```
Output: 3
```

For example:

Input	Result
1 3 4 4 2	4

Answer: (penalty regime: 0 %)

```
nums=list(map(int,input().split()))
seen=set()
for num in nums:
    if num in seen:
        print(num)
        break
    seen.add(num)
```

	Input	Expected	Got	
✓	1 3 4 4 2	4	4	✓
✓	1 2 2 3 4 5 6 7	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **10**

Correct

Mark 1.00 out of 1.00

You are given an integer tuple `nums` containing distinct numbers. Your task is to perform a sequence of operations on this tuple until it becomes empty. The operations are defined as follows:

1. If the first element of the tuple has the smallest value in the entire tuple, remove it.
2. Otherwise, move the first element to the end of the tuple.

You need to return an integer denoting the number of operations required to make the tuple empty.

Constraints

- The input tuple `nums` contains distinct integers.
- The operations must be performed using tuples and sets to maintain immutability and efficiency.
- Your function should accept the tuple `nums` as input and return the total number of operations as an integer.

Example:

Input: nums = (3, 4, -1)
Output: 5

Explanation:
Operation 1: [3, 4, -1] -> First element is not the smallest, move to the end -> [4, -1, 3]
Operation 2: [4, -1, 3] -> First element is not the smallest, move to the end -> [-1, 3, 4]
Operation 3: [-1, 3, 4] -> First element is the smallest, remove it -> [3, 4]
Operation 4: [3, 4] -> First element is the smallest, remove it -> [4]
Operation 5: [4] -> First element is the smallest, remove it -> []
Total operations: 5

For example:

Test	Result
`print(count_operations((3, 4, -1)))`	5

Answer: (penalty regime: 0 %)

Reset answer

```python
def count_operations(nums: tuple) -> int:
    # Your implementation here
    pass
    operations = 0
    remaining = set(nums)
    current = nums
    while current:
        operations += 1
        if current[0] == min(remaining):
            remaining.remove(current[0])
            current = current[1:]
        else:
            current = current[1:]+current[:1]
    return operations
```

	Test	Expected	Got	
✓	`print(count_operations((3, 4, -1)))`	5	5	✓
✓	`print(count_operations((1, 2, 3, 4, 5)))`	5	5	✓
✓	`print(count_operations((5, 4, 3, 2, 1)))`	15	15	✓
✓	`print(count_operations((42, )))`	1	1	✓
✓	`print(count_operations((-2, 3, -5, 4, 1)))`	11	11	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Status	Finished
Started	Saturday, 10 October 2026, 4:09 PM
Completed	Saturday, 10 October 2026, 4:22 PM
Duration	13 mins 36 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following are true of Python dictionaries:

a) All the keys in a dictionary must be of the same type.

b) Items are accessed by their position in a dictionary.

c) A dictionary can contain any object type except another dictionary.

d) Dictionaries can be nested to any depth.

e) Dictionaries are mutable.

f) Dictionaries are accessed by key.

a. b,c

b. a,b

c. c,d,e

d. d,e,f ✓

Question **2**

Correct

Mark 1.00 out of 1.00

What will be the output of the following Python code snippet?

```
a={}
a['a']=1
a['b']=[2,3,4]
print(a)
```

a. {'b': [2], 'a': 1}

b. ✓
{'b': [2, 3, 4], 'a': 1}

c. {'b': [2], 'a': [3]}

d. Error

Question **3**

Correct

Mark 1.00 out of 1.00

To obtain the number of entries in dictionary which command is used?

a. d.size()
b. d.len()
c. size(d)
d. len(d) ✓

Question **4**

Correct

Mark 1.00 out of 1.00

____ datatype fall under mapping.

a. List
b. Dictionary ✓
c. String
d. Tuple

Question **5**

Correct

Mark 1.00 out of 1.00

Both keys and values are unique in dictionary.

a. True
b. False ✓

Question **6**

Correct

Mark 1.00 out of 1.00

In dictionary Keys and values are separated by ____.

a. Colon (:) ✓
b. dot(.)
c. Semicolon(;)
d. Comma(,)

Question **7**

Correct

Mark 1.00 out of 1.00

clear() method is used to delete the dictionary.

a. True
b. False ✓

Question **8**

Correct

Mark 1.00 out of 1.00

The key-value pair in dictionary is called ____.

a. item ✓
b. pair item
c. value
d. paired value

Question **9**

Correct

Mark 1.00 out of 1.00

Which one of the following is correct?

a. A dictionary can have two same keys with different values.
b. A python, a dictionary can neither have two same keys nor two same values.
c. A dictionary can have two same values with different keys. ✓
d. A dictionary can have two same keys or same values but cannot have two same key-value pair

Question **10**

Correct

Mark 1.00 out of 1.00

Which function/statement delete all the items of the dictionary?

a. pop( )
b. clear( ) ✓
c. delete( )
d. del

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following are immutable data type? A. String B. Tuple C. List D. Dictionary

a. c and d
b. a and b ✓
c. b and d
d. a and c

Question **12**

Correct

Mark 1.00 out of 1.00

We can repeat the values of Key in Dictionary?

a. False
b. True ✓

Question **13**

Correct

Mark 1.00 out of 1.00

Which function helps to merge dictionary 'D1' and 'D2'?

a. update ✓
b. get
c. append
d. merge

Question **14**

Correct

Mark 1.00 out of 1.00

Dictionary is a ____ data type.

a. Ordered
b. Mapping ✓
c. Sequence
d. None of the mentioned

Question **15**

Correct

Mark 1.00 out of 1.00

Following statement return values in the form of: D1.keys() # D1 is a dictionary

a. tuple
b. list ✓
c. string
d. dictionary

Status	Finished
Started	Sunday, 11 October 2026, 12:11 PM
Completed	Sunday, 11 October 2026, 12:59 PM
Duration	47 mins 49 secs
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Objective:

Develop a Python program that takes an input string from the user and counts the number of occurrences of each vowel (a, e, i, o, u) in the string. The program should be case-insensitive, meaning it should treat uppercase and lowercase vowels as the same.

Description:

Vowels play a significant role in the English language and other alphabet-based languages. Counting vowels in a given string is a fundamental task that can be applied in various text processing applications, including speech recognition, linguistic research, and text analysis. The objective of this problem is to create a Python script that accurately counts and displays the number of times each vowel appears in a user-provided string.

Program Requirements:

Input:

First line reading String as input, The string can contain any characters, including letters, numbers, and special characters.

Output:

Display the number of occurrences of each vowel in the string.

The output should list each vowel followed by its count.

Example:

Consider the following example for better understanding:

- **Input:** "Python Programming"
- **Output**

  a = 1
  e = 0
  i = 1
  o = 2
  u = 0

For example:

Input	Result
Hello World	a = 0 e = 1 i = 0 o = 2 u = 0
Python	a = 0 e = 0 i = 0 o = 1 u = 0

Answer: (penalty regime: 0 %)

```
s=input().lower()
vowels="aeiou"
for i in vowels:
    print(i,"=",s.count(i))
```

	Input	Expected	Got	
✓	Hello World	a = 0 e = 1 i = 0 o = 2 u = 0	a = 0 e = 1 i = 0 o = 2 u = 0	✓
✓	AEIOU aeio u	a = 2 e = 2 i = 2 o = 2 u = 2	a = 2 e = 2 i = 2 o = 2 u = 2	✓
✓	Python	a = 0 e = 0 i = 0 o = 1 u = 0	a = 0 e = 0 i = 0 o = 1 u = 0	✓
✓	abcdefghijklmnopqrstuvwxyz	a = 1 e = 1 i = 1 o = 1 u = 1	a = 1 e = 1 i = 1 o = 1 u = 1	✓
✓	12345!@#$%AEIOU	a = 1 e = 1 i = 1 o = 1 u = 1	a = 1 e = 1 i = 1 o = 1 u = 1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

A company wants to send its quotation secretly to its client. The company decided to encrypt the amount they are sending to their client with some special symbols so that the equation amount will not be revealed to any external person. They used the special symbols !,@,#,$,%,^,&,*,>,< for 0,1,2,3,4,5,6,7,8,9 respectively. Write a python code to help the company to convert the amount to special symbols.

(Value rounded off to 2 decimal points)

Input

n: Float data type which reads amount to send

Output

s: : String data type which displays symbols

Sample Testcase 1

Input

10000

Output

@!!!!.!!

Sample Testcase2

1234.56

Output

@#$%.^&

For example:

Input	Result
1345.23	@$%^.#$
15000.59	@^!!!.^<
156789	@^&*><.!!

Answer: (penalty regime: 0 %)

```
n=float(input())
symbols="!@#$%^&*><"
s=""
for ch in f"{n:.2f}":
    if ch==".":
        s+="."
    else:
        s+=symbols[int(ch)]
print(s)
```

	Input	Expected	Got	
✓	1345.23	@$%^.#$	@$%^.#$	✓
✓	15000.59	@^!!!.^<	@^!!!.^<	✓
✓	1234	@#$%.!!	@#$%.!!	✓
✓	156789	@^&*><.!!	@^&*><.!!	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00

A sentence is a list of words that are separated by a single space with no leading or trailing spaces. Each word consists of lowercase and uppercase English letters.

A sentence can be shuffled by appending the 1-indexed word position to each word then rearranging the words in the sentence.

For example, the sentence "This is a sentence" can be shuffled as "sentence4 a3 is2 This1" or "is2 sentence4 This1 a3".

Given a shuffled sentence s containing no more than 9 words, reconstruct and return the original sentence.

Example 1:

Input:

is2 sentence4 This1 a3

Output:

This is a sentence

Explanation: Sort the words in s to their original positions "This1 is2 a3 sentence4", then remove the numbers.

Example 2:

Input:

Myself2 Me1 I4 and3

Output:

Me Myself and I

Explanation: Sort the words in s to their original positions "Me1 Myself2 and3 I4", then remove the numbers.

Constraints:

2 <= s.length <= 200

s consists of lowercase and uppercase English letters, spaces, and digits from 1 to 9.

The number of words in s is between 1 and 9.

The words in s are separated by a single space.

s contains no leading or trailing spaces.

Answer: (penalty regime: 0 %)

```
s=input()
words=s.split()
result=[""]*len(words)
for word in words:
    for i in word:
        if i.isdigit():
            pos=int(i)-1
            result[pos]=word.replace(i,"")
print(" ".join(result))
```

	Input	Expected	Got	
✓	is2 sentence4 This1 a3	This is a sentence	This is a sentence	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **4**

Correct

Mark 1.00 out of 1.00

A sentence is a string of single-space separated words where each word consists only of lowercase letters.A word is uncommon if it appears exactly once in one of the sentences, and does not appear in the other sentence.

Given two sentences s1 and s2, return a list of all the uncommon words. You may return the answer in any order.

Example 1:

Input: s1 = "this apple is sweet", s2 = "this apple is sour"

Output: ["sweet","sour"]

Example 2:

Input: s1 = "apple apple", s2 = "banana"

Output: ["banana"]

Constraints:

1 <= s1.length, s2.length <= 200

s1 and s2 consist of lowercase English letters and spaces.

s1 and s2 do not have leading or trailing spaces.

All the words in s1 and s2 are separated by a single space.

Note:

Use dictionary to solve the problem

For example:

Input	Result
this apple is sweet this apple is sour	sweet sour

Answer: (penalty regime: 0 %)

```
s1=input()
s2=input()
d={}
for i in s1.split():
    d[i]=d.get(i,0)+1
for i in s2.split():
    d[i]=d.get(i,0)+1
for i in d:
    if d[i]==1:
        print(i,end=" ")
```

	Input	Expected	Got	
✓	this apple is sweet this apple is sour	sweet sour	sweet sour	✓
✓	apple apple banana	banana	banana	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **5**

Correct

Mark 1.00 out of 1.00

Create a student dictionary for n students with the student name as key and their test mark assignment mark and lab mark as values. Do the following computations and display the result.

1.Identify the student with the highest average score

2.Identify the student who as the highest Assignment marks

3.Identify the student with the Lowest lab marks

4.Identify the student with the lowest average score

Note:

If more than one student has the same score display all the student names

Sample input:

4

James 67 89 56

Lalith 89 45 45

Ram 89 89 89

Sita 70 70 70

Sample Output:

Ram

James Ram

Lalith

Lalith

For example:

Input	Result
4 James 67 89 56 Lalith 89 45 45 Ram 89 89 89 Sita 70 70 70	Ram James Ram Lalith Lalith

Answer: (penalty regime: 0 %)

```
n=int(input())
students={}
for i in range(n):
    name,test,assign,lab=input().split()
    students[name]=[float(test),float(assign),float(lab)]
total={k:round(sum(v),6) for k,v in students.items()}
mx=max(total.values())
print(*sorted(k for k in total if total[k]==mx))
mx=max(v[1] for v in students.values())
print(*sorted(k for k in students if students[k][1]==mx))
mn=min(v[2] for v in students.values())
print(*sorted(k for k in students if students[k][2]==mn))
```

```
print(*sorted(k for k in students if students[k][2]==mn))
mn=min(total.values())
print(*sorted(k for k in total if total[k]==mn))
```

	Input	Expected	Got	
✓	4 James 67 89 56 Lalith 89 45 45 Ram 89 89 89 Sita 70 70 70	Ram James Ram Lalith Lalith	Ram James Ram Lalith Lalith	✓
✓	3 Raja 95 67 90 Aarav 89 90 90 ShadhanA 95 95 91	ShadhanA ShadhanA Aarav Raja Raja	ShadhanA ShadhanA Aarav Raja Raja	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **6**

Correct

Mark 1.00 out of 1.00

In the game of Scrabble™, each letter has points associated with it. The total score of a word is the sum of the scores of its letters. More common letters are worth fewer points while less common letters are worth more points. The points associated with each letter are shown below:

Points Letters

1 A, E, I, L, N, O, R, S, T and U

2 D and G

3 B, C, M and P

4 F, H, V, W and Y

5 K

8 J and X

10 Q and Z

Write a program that computes and displays the Scrabble™ score for a word. Create a dictionary that maps from letters to point values. Then use the dictionary to compute the score.

A Scrabble™ board includes some squares that multiply the value of a letter or the value of an entire word. We will ignore these squares in this exercise.

Sample Input

REC

Sample Output

REC is worth 5 points.

For example:

Input	Result
REC	REC is worth 5 points.

Answer: (penalty regime: 0 %)

```
word=input()
points={
    'A':1,'E':1,'T':1,'L':1,'N':1,
    'O':1,'R':1,'S':1,'T':1,'U':1,
    'D':2,'G':2,
    'B':3,'C':3,'M':3,'P':3,
    'F':4,'H':4,'V':4,'W':4,'Y':4,
    'K':5,
    'J':8,'X':8,
    'Q':10,'Z':10
}
score=0
for letter in word:
    score=score+points[letter]
print(word,'is worth',score,'points.')
```

	Input	Expected	Got	
✓	GOD	GOD is worth 5 points.	GOD is worth 5 points.	✓
✓	REC	REC is worth 5 points.	REC is worth 5 points.	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **7**

Correct

Mark 1.00 out of 1.00

Give a dictionary with value lists, sort the keys by summation of values in value list.

Input : test_dict = {'Gfg' : [6, 7, 4], 'best' : [7, 6, 5]}

Output : {'Gfg': 17, 'best': 18}

Explanation : Sorted by sum, and replaced.

Input : test_dict = {'Gfg' : [8,8], 'best' : [5,5]}

Output : {'best': 10, 'Gfg': 16}

Explanation : Sorted by sum, and replaced.

Sample Input:

2

Gfg 6 7 4

Best 7 6 5

Sample Output

Gfg 17

Best 18

For example:

Input	Result
2 Gfg 6 7 4 Best 7 6 5	Gfg 17 Best 18

Answer: (penalty regime: 0 %)

```
n=int(input())
d={}
for i in range(n):
    a=input().split()
    d[a[0]]=list(map(int,a[1:]))
for i in sorted(d,key=lambda x: sum(d[x])):
    print(i,sum(d[i]))
```

	Input	Expected	Got	
✓	2 Gfg 6 7 4 Best 7 6 5	Gfg 17 Best 18	Gfg 17 Best 18	✓
✓	2 Gfg 6 6 Best 5 5	Best 10 Gfg 12	Best 10 Gfg 12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **8**

Correct

Mark 1.00 out of 1.00

Given a number, convert it into corresponding alphabet.

Input	Output
1	A
26	Z
27	AA
676	YZ

Input Format

Input is an integer

Output Format

Print the alphabets

Constraints

1 <= num <= 4294967295

Sample Input 1

26

Sample Output 1

Z

For example:

Test	Result
`print(excelNumber(26))`	`Z`

Answer: (penalty regime: 0 %)

Reset answer

```
def excelNumber(n):
    s=""
    while n>0:
        n-=1
        r=n%26
        s=chr(65+r)+s
        n=n//26
    return s
```

	Test	Expected	Got	
✓	`print(excelNumber(26))`	`Z`	`Z`	✓
✓	`print(excelNumber(27))`	`AA`	`AA`	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **9**

Correct

Mark 1.00 out of 1.00

Given an array of names of candidates in an election. A candidate name in the array represents a vote cast to the candidate. Print the name of candidates received Max vote. If there is tie, print a lexicographically smaller name.

Examples:

Input : votes[] = {"john", "johnny", "jackie",
"johnny", "john", "jackie",
"jamie", "jamie", "john",
"johnny", "jamie", "johnny",
"john"};

Output : John

We have four Candidates with name as 'John', 'Johnny', 'jamie', 'jackie'. The candidates John and Johny get maximum votes. Since John is alphabetically smaller, we print it. Use dictionary to solve the above problem

Sample Input:

10
John
John
Johny
Jamie
Jamie
Johny
Jack
Johny
Johny
Jackie

Sample Output:

Johny

Answer: (penalty regime: 0 %)

```
1 n=int(input())
2 votes=[]
```

```
votes=[]
for i in range(n):
    votes.append(input())
count={}
for name in votes:
    if name in count:
        count[name]+=1
    else:
        count[name]=1
max_votes=max(count.values())
winners=[]
for name in count:
    if count[name]==max_votes:
        winners.append(name)
print(min(winners))
```

	Input	Expected	Got	
✓	10 John John Johny Jamie Jamie Johny Jack Johny Johny Jackie	Johny	Johny	✓
✓	6 Ida Ida Ida Kiruba Kiruba Kiruba	Ida	Ida	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **10**

Correct

Mark 1.00 out of 1.00

You are given a string `word`. A letter is called **special** if it appears both in lowercase and uppercase in `word`.

Your task is to return the number of **special** letters in `word`.

Constraints

- The input string `word` will contain only alphabetic characters (both lowercase and uppercase).
- The solution must utilize a dictionary to determine the number of special letters.
- The function should handle various edge cases, such as strings without any special letters, strings with only lowercase or uppercase letters, and mixed strings.

Examples

Example 1:

Input: word = "aaAbcBC"
Output: 3

Explanation:
The special characters in `word` are 'a', 'b', and 'c'.

Example 2:

Input: word = "abc"
Output: 0

Explanation:
No character in `word` appears in uppercase.

For example:

Test	Result
`print(count_special_letters("AaBbCcDdEe"))`	5

Answer: (penalty regime: 0 %)

Reset answer

```
def count_special_letters(word: str) -> int:
    # Your implementation here
    pass
    d={}
    for i in word:
        d[i.lower()]=d.get(i.lower(),0)+1
    count=0
    for i in d:
        if i in word and i.upper() in word:
            count+=1
    return count
```

	Test	Expected	Got	
✓	`print(count_special_letters("AaBbCcDdEe"))`	`5`	`5`	✓
✓	`print(count_special_letters("ABCDE"))`	`0`	`0`	✓
✓	`print(count_special_letters("abcde"))`	`0`	`0`	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Status	Finished
Started	Sunday, 18 October 2026, 8:20 PM
Completed	Sunday, 18 October 2026, 8:35 PM
Duration	15 mins 7 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(n log n)
b. O(1)
c. O👎
d. O(log n) ✓

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search is more efficient than binary search on large lists.
b. Linear search divides the list into halves.
c. Linear search can be applied to both sorted and unsorted lists. ✓
d. Linear search requires the list to be sorted.

Question **3**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Hash search
b. Binary search ✓
c. Sequential search
d. Linear search

Question **4**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search backtracks to find duplicate elements
b. The search stops immediately ✓
c. The search continues until the end of the list
d. The search starts over from the beginning

Question **5**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It compares elements sequentially
b. It can be applied only if the list is sorted ✓
c. It always starts from the beginning of the list
d. It works on unsorted lists

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Checking each element sequentially ✓
b. Dividing the list in half repeatedly
c. Sorting the list before searching
d. Skipping every second element

Question **7**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 90 and 100

b. 90 and 99 ✓

c. 89 and 94

d. 94 and 99

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. There must be a mechanism to access middle element directly

b. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

c. Binary search algorithm is not efficient when the data elements more than 1500 ✓

d. Must use a sorted array

Question **9**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is small and unsorted ✓

b. When the list is small and sorted

c. When the list is large and sorted

d. When the list is large and unsorted

Question **10**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. An error is raised
b. The search is considered unsuccessful ✓
c. The last element is returned
d. The first element is returned

Question **11**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Mining
b. Searching ✓
c. Finding
d. Discovering

Question **12**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the right sublist ✓
b. In the left sublist
c. In the middle of the list
d. At the beginning of the list

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It can only be applied to large lists
b. It does not work with negative numbers
c. It requires the list to be sorted ✓
d. It is slower than linear search for small lists

Question **14**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

- a. Binary search has a lower time complexity on large, sorted lists ✓
- b. Binary search works on unsorted lists
- c. Binary search can find multiple instances of the target element
- d. Binary search does not require dividing the list

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

- a. When the list is unsorted
- b. When the list contains duplicate elements
- c. When the list is very small
- d. When the list is sorted ✓

Status	Finished
Started	Sunday, 18 October 2026, 8:42 PM
Completed	Sunday, 18 October 2026, 9:46 PM
Duration	1 hour 3 mins
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

An list contains N numbers and you want to determine whether two of the numbers sum to a given number K. For example, if the input is 8, 4, 1, 6 and K is 10, the answer is yes (4 and 6). A number may be used twice.

Input Format

The first line contains a single integer n , the length of list

The second line contains n space-separated integers, list[i].

The third line contains integer k.

Output Format

Print Yes or No.

Sample Input

7

0 1 2 4 6 5 3

1

Sample Output

Yes

For example:

Input	Result
5 8 9 12 15 3 11	Yes
6 2 9 21 32 43 43 1 4	No

Answer: (penalty regime: 0 %)

```
n=int(input())
arr=list(map(int,input().split()))
k=int(input())
found=False
for i in range(n):
    for j in range(i+1,n):
        if arr[i]+arr[j]==k:
            found=True
            break
        if found:
            break
print("Yes" if found else "No")
```

	Input	Expected	Got	
✓	5 8 9 12 15 3 11	Yes	Yes	✓
✓	6 2 9 21 32 43 43 1 4	No	No	✓
✓	6 13 42 31 4 8 9 17	Yes	Yes	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

Given an array of integers `nums` which is sorted in ascending order, and an integer `target`, write a function to search `target` in `nums`. If `target` exists, then return its index. Otherwise, return `-1`.

You must write an algorithm with `O(log n)` runtime complexity.

Example 1:

```
Input: nums = [-1,0,3,5,9,12], target = 9
Output: 4
Explanation: 9 exists in nums and its index is 4
```

Example 2:

```
Input: nums = [-1,0,3,5,9,12], target = 2
Output: -1
Explanation: 2 does not exist in nums so return -1
```

Constraints:

- `1 <= nums.length <= 10^4`
- `-10^4 < nums[i], target < 10^4`
- All the integers in `nums` are **unique**.
- `nums` is sorted in ascending order.

For example:

Test	Result
`print(search([-1,0,3,5,9,12],9))`	4

Answer: (penalty regime: 0 %)

Reset answer

```
def search(nums,target):
    left,right=0,len(nums)-1
    while left<=right:
        mid=(left+right)//2
        if nums[mid]==target:
            return mid
        elif nums[mid]<target:
            left=mid+1
        else:
            right=mid-1
    return -1

```

	Test	Expected	Got	
✓	`print(search([-1,0,3,5,9,12],9))`	`4`	`4`	✓
✓	`print(search([-1,0,3,5,9,12],2))`	`-1`	`-1`	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00

Given two Strings s1 and s2, remove all the characters from s1 which is present in s2.

Constraints

1<= string length <= 200

Sample Input 1

experience
enc

Sample Output 1

xpri

Answer: (penalty regime: 0 %)

```
1 s1=input()
2 s2=input()
3 print("".join(c for c in s1 if c not in s2))
```

	Input	Expected	Got	
✓	experience enc	xpri	xpri	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **4**

Correct

Mark 1.00 out of 1.00

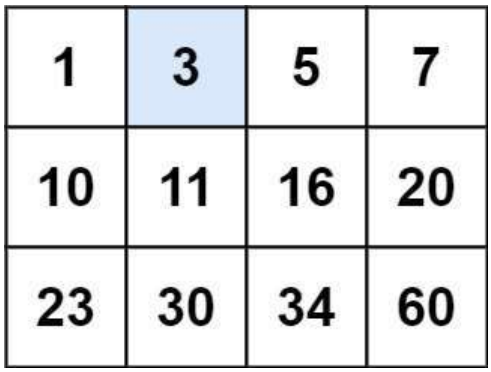
You are given an `m x n` integer matrix `matrix` with the following two properties:

- Each row is sorted in non-decreasing order.
- The first integer of each row is greater than the last integer of the previous row.

Given an integer `target`, return `True` *if* `target` *is in* `matrix` *or* `False` *otherwise*.

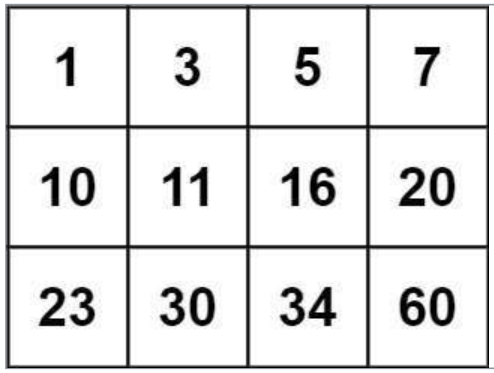
You must write a solution in `O(log(m * n))` time complexity.

Example 1:




```
Input: matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]], target = 3
Output: True
```

Example 2:




```
Input: matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]], target = 13
Output: False
```

For example:

Test	Result
print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 13))	False
print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 3))	True

Answer: (penalty regime: 0 %)

Reset answer

```
from typing import List
def searchMatrix(matrix: List[List[int]], target: int) -> bool:
    m,n=len(matrix),len(matrix[0])
    left,right=0,m*n-1
    while left<=right:
```

```
        mid=(left+right)//2
        val=matrix[mid//n][mid%n]
        if val==target:
            return True
        elif val<target:
            left=mid+1
        else:
            right=mid-1
    return False

```

	Test	Expected	Got	
✓	print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 13))	False	False	✓
✓	print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 3))	True	True	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **5**

Correct

Mark 1.00 out of 1.00

Two string values S1, S2 are passed as the input. The program must print first N characters present in S1 which are also present in S2.

Input Format:

The first line contains S1.
The second line contains S2.
The third line contains N.

Output Format:

The first line contains the N characters present in S1 which are also present in S2.

Boundary Conditions:

2 <= N <= 10
2 <= Length of S1, S2 <= 1000

Example Input/Output 1:

Input:

abcbde
cdefghbb
3

Output:

bcd

Note:

b occurs twice in common but must be printed only once.

Answer: (penalty regime: 0 %)

```
s1=input()
s2=input()
n=int(input())
res=[]
for c in s1:
    if c in s1:
        if c in s2 and c not in res:
            res.append(c)
print("".join(res[:n]))
```

	Input	Expected	Got	
✓	abcbde cdefghbb 3	bcd	bcd	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **6**

Correct

Mark 1.00 out of 1.00

Balanced strings are those that have an equal quantity of 'L' and 'R' characters.

Given a balanced string s, split it in the maximum amount of balanced strings.

Return the maximum amount of split balanced strings.

Example 1:

Input:

RLRRLLRLRL

Output:

4

Explanation: s can be split into "RL", "RRLL", "RL", "RL", each substring contains same number of 'L' and 'R'.

Example 2:

Input:

RLLLLRRRLR

Output:

3

Explanation: s can be split into "RL", "LLLRRR", "LR", each substring contains same number of 'L' and 'R'.

Example 3:

Input:

LLLLRRRR

Output:

1

Explanation: s can be split into "LLLLRRRR".

Constraints:

1 <= s.length <= 1000

s[i] is either 'L' or 'R'.

s is a balanced string.

For example:

Test	Result
`print(BalancedStrings('RLRRLLRLRL'))`	4
`print(BalancedStrings('RLLLLRRRLR'))`	3

Answer: (penalty regime: 0 %)

Reset answer

```
def BalancedStrings(s):
    count=bal=0
    for ch in s:
        bal+=1 if ch=='L' else -1
        if bal==0:
            count+=1
    return count
```

	Test	Expected	Got	
✓	`print(BalancedStrings('RLRRLLRLRL'))`	4	4	✓
✓	`print(BalancedStrings('RLLLLRRRLR'))`	3	3	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **7**

Correct

Mark 1.00 out of 1.00

Write a Python program for binary search.

For example:

Input	Result
1,2,3,5,8 6	False
3,5,9,45,42 42	True

Answer: (penalty regime: 0 %)

```
arr=sorted(map(int,input().split(",")))
target=int(input())
left,right=0,len(arr)-1
found=False
while left<=right:
    mid=(left+right)//2
    if arr[mid]==target:
        found=True
        break
    elif arr[mid]<target:
        left=mid+1
    else:
        right=mid-1
print(found)
```

	Input	Expected	Got	
✓	1,2,3,5,8 6	False	False	✓
✓	3,5,9,45,42 42	True	True	✓
✓	52,45,89,43,11 11	True	True	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **8**

Correct

Mark 1.00 out of 1.00

String should contain only the words are not palindrome.

Sample Input 1

Malayalam is my mother tongue

Sample Output 1

is my mother tongue

Answer: (penalty regime: 0 %)

```
s=input().lower()
print(" ".join(w for w in s.split() if w!=w[::-1]))
```

	Input	Expected	Got	
✓	Malayalam is my mother tongue	is my mother tongue	is my mother tongue	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **9**

Correct

Mark 1.00 out of 1.00

Given an array `nums` containing `n` distinct numbers in the range `[0, n]`, return *the only number in the range that is missing from the array.*

Example 1:

```
Input: nums = [3,0,1]
Output: 2
Explanation: n = 3 since there are 3 numbers, so all numbers are in the range [0,3]. 2 is the missing number in the range since it does not appear in nums.
```

Example 2:

```
Input: nums = [0,1]
Output: 2
Explanation: n = 2 since there are 2 numbers, so all numbers are in the range [0,2]. 2 is the missing number in the range since it does not appear in nums.
```

Example 3:

```
Input: nums = [9,6,4,2,3,5,7,0,1]
Output: 8
Explanation: n = 9 since there are 9 numbers, so all numbers are in the range [0,9]. 8 is the missing number in the range since it does not appear in nums.
```

For example:

Test	Result
`print(missingNumber([3,0,1]))`	2
`print(missingNumber([0,1]))`	2

Answer: (penalty regime: 0 %)

Reset answer

```
def missingNumber(numbers):
    n=len(numbers)
    return n*(n+1)//2-sum(numbers)
```

	Test	Expected	Got	
✓	`print(missingNumber([3,0,1]))`	2	2	✓
✓	`print(missingNumber([0,1]))`	2	2	✓
✓	`print(missingNumber([9,6,4,2,3,5,7,0,1]))`	8	8	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **10**

Correct

Mark 1.00 out of 1.00

Given an list, find peak element in it. A peak element is an element that is greater than its neighbors.

An element a[i] is a peak element if

A[i-1] <= A[i] >=a[i+1] for middle elements. [0<i<n-1]

A[i-1] <= A[i] for last element [i=n-1]

A[i]>=A[i+1] for first element [i=0]

Input Format

The first line contains a single integer n , the length of A .
The second line contains n space-separated integers,A[i].

Output Format

Print peak numbers separated by space.

Sample Input

5

8 9 10 2 6

Sample Output

```
10 6
```

For example:

Input	Result
4 12 3 6 8	12 8

Answer: (penalty regime: 0 %)

```
n=int(input())
a=list(map(int,input().split()))
res=[]
for i in range(n):
    left_ok=(i==0) or a[i-1]<=a[i]
    right_ok=(i==n-1) or a[i]>=a[i+1]
    if left_ok and right_ok:
        res.append(a[i])
print(*res)
```

	Input	Expected	Got	
✓	7 15 7 10 8 9 4 6	15 10 9 6	15 10 9 6	✓
✓	4 12 3 6 8	12 8	12 8	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Status	Finished
Started	Monday, 19 October 2026, 7:06 PM
Completed	Monday, 19 October 2026, 7:14 PM
Duration	8 mins 9 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Arranging data in a specific order
- b. Finding a specific element in a list
- c. Removing duplicates from a list
- d. Merging two datasets

The correct answer is: Arranging data in a specific order

Question **2**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Makes data harder to manage
- b. Provides a basis for other algorithms to work efficiently
- c. Makes code execution slower
- d. Decreases the efficiency of algorithms

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **3**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Binary Search
b. Linear Search
c. Bubble Sort
d. Merge Sort

The correct answer is: Merge Sort

Question **4**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

a. A sorting algorithm is stable if it preserves the order of all keys
b. A sorting algorithm is stable if it preserves the order of non-duplicate keys
c. A sorting algorithm is stable if it preserves the order of duplicate keys
d. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **5**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It slows down the process
b. It complicates the selection of items
c. It makes the data unsorted
d. It makes it easier to select items based on their relationship to the rest of the items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **6**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 40,70,80,90,120,150,200,300,400,470

b. 200,470,80,150,40,90,300,400,70,120

c. 40,80,90,150,200,300,400,470,70,120

d. 80,150,200,470,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **7**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Bubble Sort

b. Merge Sort

c. Heap Sort

d. Quick Sort

The correct answer is: Bubble Sort

Question **8**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Bubble Sort

b. Quick Sort

c. Merge Sort

d. Insertion Sort

The correct answer is: Bubble Sort

Question **9**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. order()
b. sorted()
c. arrange()
d. sort()

The correct answer is: sort()

Question **10**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Heap sort
b. Insertion sort
c. Bubble sort
d. Quick sort

The correct answer is:
Insertion sort

Question **11**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. The parts are ignored
b. Each part is searched for a specific element
c. The parts are merged without sorting
d. Each part is sorted independently

The correct answer is: Each part is sorted independently

Question **12**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Linear Search
b. Quick Sort
c. Bubble Sort
d. Merge Sort

The correct answer is: Merge Sort

Question **13**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. Duplicates
b. Searching
c. Selection
d. All of the above

The correct answer is: All of the above

Question **14**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Dividing the input into smaller subproblems
b. Sorting the entire list sequentially
c. Comparing each element with the others
d. Combining sorted sublists

The correct answer is: Dividing the input into smaller subproblems

Question **15**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By comparing elements sequentially
b. By sorting data in a single pass
c. By breaking the input into smaller parts and merging them
d. By using the bubble-up method

The correct answer is: By breaking the input into smaller parts and merging them

Status	Finished
Started	Monday, 19 October 2026, 7:20 PM
Completed	Monday, 19 October 2026, 8:30 PM
Duration	1 hour 9 mins
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

To find the frequency of numbers in a list and display in sorted order.

Constraints:

1<=n, arr[i]<=100

Input:

1 68 79 4 90 68 1 4 5

output:

1 2

4 2

5 1

68 2

79 1

90 1

For example:

Input	Result
4 3 5 3 4 5	3 2 4 2 5 2

Answer: (penalty regime: 0 %)

```
from collections import Counter
arr=list(map(int,input().split()))
for k,v in sorted(Counter(arr).items()):
    print(k,v)
```

	Input	Expected	Got	
✓	4 3 5 3 4 5	3 2 4 2 5 2	3 2 4 2 5 2	✓

	Input	Expected	Got	
✓	12 4 4 4 2 3 5	2 1 3 1 4 3 5 1 12 1	2 1 3 1 4 3 5 1 12 1	✓
✓	5 4 5 4 6 5 7 3	3 1 4 2 5 3 6 1 7 1	3 1 4 2 5 3 6 1 7 1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

Given an integer array `nums`, return *an integer array* `counts` *where* `counts[i]` *is the number of smaller elements to the right of* `nums[i]`.

Example 1:

```
Input: nums = [5,2,6,1]
Output: [2,1,1,0]
Explanation:
To the right of 5 there are 2 smaller elements (2 and 1).
To the right of 2 there is only 1 smaller element (1).
To the right of 6 there is 1 smaller element (1).
To the right of 1 there is 0 smaller element.
```

Example 2:

```
Input: nums = [-1]
Output: [0]
```

Example 3:

```
Input: nums = [-1,-1]
Output: [0,0]
```

Constraints:

- `1 <= nums.length <= 10^5`
- `-10^4 <= nums[i] <= 10^4`

For example:

Test	Result
`print(countSmaller([5,2,6,1]))`	`[2, 1, 1, 0]`
`print(countSmaller([-1]))`	`[0]`

Answer: (penalty regime: 0 %)

Reset answer

```
from typing import List
def countSmaller(nums: List[int]) -> List[int]:
    OFFSET=10001
    size=20002
    tree=[0]*(size+1)
    def update(i):
        while i<=size:
            tree[i]+=1
            i+=i&-i
    def query(i):
        s=0
        while i>0:
            s+=tree[i]
            i=i-(i&-i)
        return s
    res=[0]*len(nums)
    for idx in range(len(nums)-1,-1,-1):
        pos=nums[idx]+OFFSET
        res[idx]=query(pos-1)
        update(pos)
    return res

```

	Test	Expected	Got	
✓	`print(countSmaller([5,2,6,1]))`	`[2, 1, 1, 0]`	`[2, 1, 1, 0]`	✓
✓	`print(countSmaller([50,20,60,10]))`	`[2, 1, 1, 0]`	`[2, 1, 1, 0]`	✓
✓	`print(countSmaller([-1]))`	`[0]`	`[0]`	✓
✓	`print(countSmaller([-1, -1]))`	`[0, 0]`	`[0, 0]`	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00

Objective:

Develop a Python program to find the k-th maximum value in a given list of integers. The program should handle various edge cases, including lists with duplicate values, empty lists, and invalid values of k. The k-th maximum value refers to the k-th largest distinct element in the list.

Background:

Finding the k-th maximum value in a list is a common problem in computer science, often encountered in fields like data analysis, competitive programming, and software development. This problem requires an understanding of sorting algorithms, data structures, and efficient problem-solving techniques. By solving this problem, one gains insights into how to handle large datasets and optimize performance in practical applications.

Problem Description:

Given a list of integers, the task is to determine the k-th maximum value in the list. The program should meet the following requirements:

1. **Input:**
    - A list of integers, which can contain both positive and negative values.
    - An integer k, representing the position of the maximum value to find.
2. **Output:**
    - The k-th maximum value in the list.
    - If k is greater than the number of distinct elements in the list or if the list is empty, the program should return an appropriate message indicating the error.

Constraints:

- The list may contain duplicate values.
- The value of k should be a positive integer.
- The list may contain up to 10^6 elements, and each element can be as large as 10^9 in magnitude.

Examples:

Consider the following examples for better understanding:

1. **Example 1:**
    - **Input:** list = [3, 1, 5, 4, 2], k = 2
    - **Output:** 4
    - **Explanation:** The distinct elements in the list are [1, 2, 3, 4, 5]. The 2nd maximum value is 4.
2. **Example 2:**
    - **Input:** list = [7, 7, 7, 7, 7], k = 1
    - **Output:** 7
    - **Explanation:** The distinct elements in the list are [7]. The 1st maximum value is 7.
3. **Example 3:**
    - **Input:** list = [2, 1, 2, 1, 2], k = 3
    - **Output:** -1
    - **Explanation:** The distinct elements in the list are [1, 2]. There is no 3rd maximum value.

For example:

Input	Result
5 3 1 5 4 2 2	4

Input	Result
6 7 7 7 7 7 7 1	7
10 2 1 2 1 2 1 2 1 2 1 3	-1

Answer: (penalty regime: 0 %)

```
n=int(input())
lst=list(map(int,input().split()))
k=int(input())
distinct=sorted(set(lst),reverse=True)
if not lst or k<1 or k>len(distinct):
    print(-1)
else:
    print(distinct[k-1])
```

	Input	Expected	Got	
✓	5 3 1 5 4 2 2	4	4	✓
✓	6 7 7 7 7 7 7 1	7	7	✓
✓	10 2 1 2 1 2 1 2 1 2 1 3	-1	-1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **4**

Correct

Mark 1.00 out of 1.00

Bubble Sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements if they are in wrong order. You read an list of numbers. You need to arrange the elements in ascending order and print the result. The sorting should be done using bubble sort.

Input Format: The first line reads the number of elements in the array. The second line reads the array elements one by one.

Output Format: The output should be a sorted list.

For example:

Input	Result
6 3 4 8 7 1 2	1 2 3 4 7 8
5 4 5 2 3 1	1 2 3 4 5

Answer: (penalty regime: 0 %)

```
n=int(input())
a=list(map(int,input().split()))
for i in range(n):
    for j in range(n-i-1):
        if a[j]>a[j+1]:
            a[j],a[j+1]=a[j+1],a[j]
print(*a)
```

	Input	Expected	Got	
✓	6 3 4 8 7 1 2	1 2 3 4 7 8	1 2 3 4 7 8	✓
✓	6 9 18 1 3 4 6	1 3 4 6 9 18	1 3 4 6 9 18	✓
✓	5 4 5 2 3 1	1 2 3 4 5	1 2 3 4 5	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **5**

Correct

Mark 1.00 out of 1.00

Given an listof integers, sort the array in ascending order using the *Bubble Sort* algorithm above. Once sorted, print the following three lines:

1. List is sorted in numSwaps swaps., where numSwaps is the number of swaps that took place.
2. First Element: firstElement, the *first* element in the sorted list.
3. Last Element: lastElement, the *last* element in the sorted list.

For example, given a worst-case but small array to sort: a=[6,4,1]. It took 3 swaps to sort the array. Output would be

```
Array is sorted in 3 swaps.
```

```
First Element: 1
```

```
Last Element: 6
```

Input Format

The first line contains an integer,n , the size of the list a .
The second line contains n, space-separated integers a[i].

Constraints

· 2<=n<=600

· 1<=a[i]<=$2x10^6$.

Output Format

You must print the following three lines of output:

1. List is sorted in numSwaps swaps., where numSwaps is the number of swaps that took place.
2. First Element: firstElement, the *first* element in the sorted list.
3. Last Element: lastElement, the *last* element in the sorted list.

Sample Input 0

3

1 2 3

Sample Output 0

List is sorted in 0 swaps.

First Element: 1

Last Element: 3

For example:

Input	Result
3 3 2 1	List is sorted in 3 swaps. First Element: 1 Last Element: 3
5 1 9 2 8 4	List is sorted in 4 swaps. First Element: 1 Last Element: 9

Answer: (penalty regime: 0 %)

```
n=int(input())
a=list(map(int,input().split()))
swaps=0
```

```
swaps=0
for i in range(n):
    for j in range(n-i-1):
        if a[j]>a[j+1]:
            a[j],a[j+1]=a[j+1],a[j]
            swaps+=1
print("List is sorted in",swaps,"swaps.")
print("First Element:",a[0])
print("Last Element:",a[-1])
```

	Input	Expected	Got	
✓	3 3 2 1	List is sorted in 3 swaps. First Element: 1 Last Element: 3	List is sorted in 3 swaps. First Element: 1 Last Element: 3	✓
✓	5 1 9 2 8 4	List is sorted in 4 swaps. First Element: 1 Last Element: 9	List is sorted in 4 swaps. First Element: 1 Last Element: 9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **6**

Correct

Mark 1.00 out of 1.00

Given an array of integers `arr`, replace each element with its rank.

The rank represents how large the element is. The rank has the following rules:

- Rank is an integer starting from 1.
- The larger the element, the larger the rank. If two elements are equal, their rank must be the same.
- Rank should be as small as possible.

Example 1:

```
Input: arr = [40,10,20,30]
Output: [4,1,2,3]
Explanation: 40 is the largest element. 10 is the smallest. 20 is the second smallest. 30 is the third smallest.
```

Example 2:

```
Input: arr = [100,100,100]
Output: [1,1,1]
Explanation: Same elements share the same rank.
```

Example 3:

```
Input: arr = [37,12,28,9,100,56,80,5,12]
Output: [5,3,4,2,8,6,7,1,3]
```

Constraints:

- `0 <= arr.length <= 10^5`
- `-10^9 <= arr[i] <= 10^9`

For example:

Test	Result
`print(arrayRankTransform([40,10,20,30]))`	`[4, 1, 2, 3]`

Answer: (penalty regime: 0 %)

Reset answer

```
from typing import List
def arrayRankTransform(arr: List[int]) -> List[int]:
    rank={v:i+1 for i,v in enumerate(sorted(set(arr)))}
    return [rank[x] for x in arr]


```

	Test	Expected	Got	
✓	`print(arrayRankTransform([40,10,20,30]))`	`[4, 1, 2, 3]`	`[4, 1, 2, 3]`	✓
✓	`print(arrayRankTransform([100,100,100]))`	`[1, 1, 1]`	`[1, 1, 1]`	✓
✓	`print(arrayRankTransform([37,12,28,9,100,56,80,5,12]))`	`[5, 3, 4, 2, 8, 6, 7, 1, 3]`	`[5, 3, 4, 2, 8, 6, 7, 1, 3]`	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **7**

Correct

Mark 1.00 out of 1.00

Write a Python program to sort a list of elements using the merge sort algorithm.

For example:

Input	Result
5 6 5 4 3 8	3 4 5 6 8

Answer: (penalty regime: 0 %)

```
def merge(a):
    if len(a)<=1:
        return a
    mid=len(a)//2
    left=merge(a[:mid])
    right=merge(a[mid:])
    res=[]
    i=j=0
    while i<len(left) and j<len(right):
        if left[i]<=right[j]:
            res.append(left[i])
            i+=1
        else:
            res.append(right[j])
            j+=1
    res.extend(left[i:])
    res.extend(right[j:])
    return res
n=int(input())
a=list(map(int,input().split()))
print(*merge(a))
```

	Input	Expected	Got	
✓	5 6 5 4 3 8	3 4 5 6 8	3 4 5 6 8	✓
✓	9 14 46 43 27 57 41 45 21 70	14 21 27 41 43 45 46 57 70	14 21 27 41 43 45 46 57 70	✓
✓	4 86 43 23 49	23 43 49 86	23 43 49 86	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **8**

Correct

Mark 1.00 out of 1.00

Given an integer array `nums` sorted in **non-decreasing** order, return *an array of **the squares of each number** sorted in non-decreasing order.*

Example 1:

```
Input: nums = [-4,-1,0,3,10]
Output: [0,1,9,16,100]
Explanation: After squaring, the array becomes [16,1,0,9,100].
After sorting, it becomes [0,1,9,16,100].
```

Example 2:

```
Input: nums = [-7,-3,2,3,11]
Output: [4,9,9,49,121]
```

Constraints:

- `1 <= nums.length <= 10^4`
- `-10^4 <= nums[i] <= 10^4`
- `nums` is sorted in **non-decreasing** order.

For example:

Test	Result
`print(sortedSquares([-4,-1,0,3,10]))`	`[0, 1, 9, 16, 100]`

Answer: (penalty regime: 0 %)

Reset answer

```python
from typing import List
def sortedSquares(nums: List[int]) -> List[int]:
    n=len(nums)
    res=[0]*n
    l,r=0,n-1
    for p in range(n-1,-1,-1):
        if abs(nums[l])>abs(nums[r]):
            res[p]=nums[l]**2
            l+=1
        else:
            res[p]=nums[r]**2
            r-=1
    return res

```

	Test	Expected	Got	
✓	`print(sortedSquares([-4,-1,0,3,10]))`	`[0, 1, 9, 16, 100]`	`[0, 1, 9, 16, 100]`	✓

	Test	Expected	Got	
✓	`print(sortedSquares([-7,-3,2,3,11]))`	`[4, 9, 9, 49, 121]`	`[4, 9, 9, 49, 121]`	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **9**

Correct

Mark 1.00 out of 1.00

Given an array of integers `nums`, sort the array in **increasing** order based on the frequency of the values. If multiple values have the same frequency, sort them in **decreasing** order.

Print the *sorted array*.

Example 1:

```
Input:
6
1 1 2 2 2 3

Output:
3 1 1 2 2 2

Explanation: '3' has a frequency of 1, '1' has a frequency of 2, and '2' has a frequency of 3.
```

Example 2:

```
Input:
5
2 3 1 3 2
Output:
1 3 3 2 2
Explanation: '2' and '3' both have a frequency of 2, so they are sorted in decreasing order.
```

Example 3:

```
Input:
9
-1 1 -6 4 5 -6 1 4 1
Output:
5 -1 4 4 -6 -6 1 1 1
```

Constraints:

- `1 <= nums.length <= 100`
- `-100 <= nums[i] <= 100`

For example:

Input	Result
6 1 1 2 2 2 3	3 1 1 2 2 2
5 2 3 1 3 2	1 3 3 2 2

Answer: (penalty regime: 0 %)

```
from collections import Counter
n=int(input())
nums=list(map(int,input().split()))
c=Counter(nums)
print(*sorted(nums,key=lambda x: (c[x], -x)))
```

	Input	Expected	Got	
✓	6 1 1 2 2 2 3	3 1 1 2 2 2	3 1 1 2 2 2	✓
✓	5 2 3 1 3 2	1 3 3 2 2	1 3 3 2 2	✓
✓	9 -1 1 -6 4 5 -6 1 4 1	5 -1 4 4 -6 -6 1 1 1	5 -1 4 4 -6 -6 1 1 1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **10**

Correct

Mark 1.00 out of 1.00

The problem is that we want to reverse a array in O(N) linear time complexity and we want the algorithm to be in-place as well!
For example: input is [1,2,3,4,5] then the output is [5,4,3,2,1]
Input
5
1 2 3 4 5
Output
5 4 3 2 1

For example:

Input	Result
5 1 2 3 4 5	5 4 3 2 1

Answer: (penalty regime: 0 %)

```
n=int(input())
a=list(map(int,input().split()))
i,j=0,n-1
while i<j:
    a[i],a[j]=a[j],a[i]
    i+=1
    j-=1
print(*a)
```

	Input	Expected	Got	
✓	5 1 2 3 4 5	5 4 3 2 1	5 4 3 2 1	✓
✓	10 0 2 4 6 8 1 3 5 7 9	9 7 5 3 1 8 6 4 2 0	9 7 5 3 1 8 6 4 2 0	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Status	Finished
Started	Monday, 19 October 2026, 8:33 PM
Completed	Monday, 19 October 2026, 8:37 PM
Duration	4 mins 8 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the purpose of the with statement in Python file handling?

a. To delete a file
b. To automatically close the file after its suite finishes ✓
c. To write data to a file
d. To read data from a file

The correct answer is: To automatically close the file after its suite finishes

Question **2**

Correct

Mark 1.00 out of 1.00

What will the read() method return if used on an empty file?

a. An empty string ✓
b. 0
c. None
d. EOFError

The correct answer is: An empty string

Question **3**

Correct

Mark 1.00 out of 1.00

Default EOL character of text file in python is ,

a. \t

b. ,

c. space

d. \n ✓

The correct answer is:
\n

Question **4**

Correct

Mark 1.00 out of 1.00

What does the following code do?

```
f = open("data.txt", "a")
f.write("Adding new data\n")
f.close()
```

a. Overwrites the file with "Adding new data"

b. Appends "Adding new data" to the end of the file ✓

c. Opens the file in read mode

d. Raises an error because the file is opened in append mode

The correct answer is: Appends "Adding new data" to the end of the file

Question **5**

Correct

Mark 1.00 out of 1.00

What does the flush() method do in file handling?

a. Moves the file cursor to the beginning

b. Deletes the file content

c. Closes the file

d. Flushes the internal buffer to the file ✓

The correct answer is: Flushes the internal buffer to the file

Question **6**

Correct

Mark 1.00 out of 1.00

Which method is used to read a specific number of characters from a file?

a. readchar()
b. readline()
c. read(size) ✓
d. readlines()

The correct answer is: read(size)

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following modes opens a file for both reading and writing?

a. r+ ✓
b. a
c. w
d. r

The correct answer is: r+

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following modes opens a file for writing in binary format?

a. w
b. r+
c. wb ✓
d. rb

The correct answer is: wb

Question **9**

Correct

Mark 1.00 out of 1.00

Which method is used to sets the position of a file pointer

a. fseek()
b. seek() ✓
c. ftell()
d. tell()

The correct answer is: seek()

Question **10**

Correct

Mark 1.00 out of 1.00

What does the mode attribute of a file object indicate?

a. The size of the file
b. The access mode in which the file was opened ✓
c. The current position of the file cursor
d. The name of the file

The correct answer is: The access mode in which the file was opened

Question **11**

Correct

Mark 1.00 out of 1.00

Select the correct method to write a list of lines to a file

a. writelist(list)
b. write(list)
c. writelines(list) ✓
d. None of the mentioned

The correct answer is: writelines(list)

Question **12**

Correct

Mark 1.00 out of 1.00

What is the default mode in which the open() function opens a file?

a. Write
b. Read ✓
c. Binary
d. Append

The correct answer is: Read

Question **13**

Correct

Mark 1.00 out of 1.00

How do you check if a file is closed in Python?

a. file.is_closed()
b. file.closed ✓
c. file.isclosed()
d. file.closed()

The correct answer is: file.closed

Question **14**

Correct

Mark 1.00 out of 1.00

Select the incorrect file access mode

a. r
b. ab+
c. wb+
d. rw+ ✓

The correct answer is: rw+

Question **15**

Correct

Mark 1.00 out of 1.00

Which method is used to truncate a file to a specific size?

a. shrink()
b. resize()
c. reduce()
d. truncate() ✓

The correct answer is: truncate()

Status	Finished
Started	Monday, 19 October 2026, 8:42 PM
Completed	Monday, 19 October 2026, 9:36 PM
Duration	53 mins 56 secs
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Create a Python program to write to a specific line in a text file, replacing the existing content of that line.

Description:

1. **Input:**
   - A text file with multiple lines.
   - A line number to write to.
   - New content for the specified line.
2. **Output:**
   - The updated file with the specified line replaced by the new content in file "output.txt".

Example:

- **Input File Content:**

  "Line one.
  Line two.
  Line three.
  Line four."
  2

  Updated line two.

Output:

Line one.
Updated line two.
Line three.
Line four.

For example:

Test	Input	Result

`with open('output.txt', 'r') as file:`
`  text = file.read()`
`  print(text)`
	`input1.txt`
	`2`
	`Updated line two.`
		`Line one.`
		`Updated line two.`
		`Line three.`
		`Line four.`

Answer: (penalty regime: 0 %)

```
name=input()
n=int(input())
new=input()
with open(name) as f:
    lines=f.read().splitlines()
if 1<=n<=len(lines):
    lines[n-1]=new
with open('output.txt','w') as f:
    f.write('\n'.join(lines)+'\n')
```

	Test	Input	Expected	Got	

✓	`with open('output.txt', 'r') as file:`
	`  text = file.read()`
	`  print(text)`
		`input1.txt`
		`2`
		`Updated line two.`
			`Line one.`
			`Updated line two.`
			`Line three.`
			`Line four.`
				`Line one.`
				`Updated line two.`
				`Line three.`
				`Line four.`
					✓

✓	`with open('output.txt', 'r') as file:`
	`  text = file.read()`
	`  print(text)`
		`input2.txt`
		`2`
		`Line B Updated.`
			`Line A.`
			`Line B Updated.`
			`Line C.`
				`Line A.`
				`Line B Updated.`
				`Line C.`
					✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

Create a Python program to find the longest word in a text file.

- **Input:**
  - A text file containing multiple lines of text.
- **Output:**
  - The longest word in the file.

For example:

Input	Result
input1.txt	Longest word: containing

Answer: (penalty regime: 0 %)

```
name=input()
with open(name) as f:
    words=f.read().split()
print("Longest word:",max(words,key=len))
```

	Input	Expected	Got	
✓	input1.txt	Longest word: containing	Longest word: containing	✓
✓	input2.txt	Longest word: thousand	Longest word: thousand	✓
✓	input3.txt	Longest word: supercalifragilisticexpialidocious	Longest word: supercalifragilisticexpialidocious	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00

Write a Python program to count the frequency of each word in a given text file.

Description:

1. **Input:**
   - String as input.
2. **Output:**
   - A list of words with their corresponding frequency count to be write in a file "output.txt"

Example:

- **Input File Content:**

  apple orange apple banana apple orange

Output:

apple: 3
orange: 2
banana: 1

For example:

Test	Input	Result

`with open('output.txt', 'r') as file:`
`  text = file.read()`
`  print(text)`
	`apple orange apple banana apple orange`	`apple: 3`
		`banana: 1`
		`orange: 2`

Answer: (penalty regime: 0 %)

```
text=input()
counts={}
for w in text.split():
    w=w.strip('.,!?;:"\'()').lower()
    if w:
        counts[w]=counts.get(w,0)+1
with open('output.txt','w') as f:
    for w in sorted(counts):
        f.write(f"{w}: {counts[w]}\n")
```

	Test	Input	Expected	Got	

✓	`with open('output.txt', 'r')`
	`as file:`
	`  text = file.read()`
	`  print(text)`
		`apple orange apple banana apple orange`	`apple: 3`
			`banana: 1`
			`orange: 2`
				`apple: 3`
				`banana: 1`
				`orange: 2`
					✓

	Test	Input	Expected	Got	

✓	`with open('output.txt', 'r') as file:`
	`  text = file.read()`
	`  print(text)`
		`Hello world! Hello everyone. Welcome to the world of programming.`	`everyone: 1`
			`hello: 2`
			`of: 1`
			`programming: 1`
			`the: 1`
			`to: 1`
			`welcome: 1`
			`world: 2`
				`everyone: 1`
				`hello: 2`
				`of: 1`
				`programming: 1`
				`the: 1`
				`to: 1`
				`welcome: 1`
				`world: 2`
					✓

✓	`with open('output.txt', 'r') as file:`
	`  text = file.read()`
	`  print(text)`
		`One fish two fish Red fish blue fish`	`blue: 1`
			`fish: 4`
			`one: 1`
			`red: 1`
			`two: 1`
				`blue: 1`
				`fish: 4`
				`one: 1`
				`red: 1`
				`two: 1`
					✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **4**

Correct

Mark 1.00 out of 1.00

Write a Python program to reverse the contents of a specific line in a text file based on a given line number.

Description:

1. **Input:**
   - A text file with multiple lines.
   - A line number to reverse.
2. **Output:**
   - The updated file with the specified line's contents reversed in file "output.txt".

Example:

- **Input File Content:**

  "Line one.
  Line two.
  Line three.
  Line four."
  3

Output:

Line one.
Line two.
eerht eniL.
Line four.

For example:

Test	Input	Result

`with open('output.txt', 'r') as file:`
`  text = file.read()`
`  print(text)`
	`input1.txt`
	`3`
		`Line one.`
		`Line two.`
		`eerht eniL.`
		`Line four.`

Answer: (penalty regime: 0 %)

```
name=input()
n=int(input())
with open(name) as f:
    lines=f.read().splitlines()
if 1<=n<=len(lines):
    line=lines[n-1]
    if line.endswith('.'):
        lines[n-1]=line[:-1][::-1]+'.'
with open('output.txt','w') as f:
    f.write('\n'.join(lines)+'\n')
```

	Test	Input	Expected	Got	

✓	`with open('output.txt', 'r') as file:`
	`  text = file.read()`
	`  print(text)`
		input1.txt
		3
			Line one.
			Line two.
			eerht eniL.
			Line four.
				Line one.
				Line two.
				eerht eniL.
				Line four.
					✓

✓	`with open('output.txt', 'r') as file:`
	`  text = file.read()`
	`  print(text)`
		input2.txt
		2
			Line A.
			B eniL.
			Line C.
				Line A.
				B eniL.
				Line C.
					✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **5**

Correct

Mark 1.00 out of 1.00

Develop a Python program to identify and print all palindrome words from a given text file.

Description:

1. **Input:**
   - A text file containing multiple words.
2. **Output:**
   - A list of palindrome words found in the file name as 'output.txt'.

For example:

Test	Input	Result

`with open('output.txt', 'r') as file:`
`  text = file.read()`
`  print(text)`
	input1.txt	madam
		arora
		malayalam

Answer: (penalty regime: 0 %)

```
name=input()
with open(name) as f:
    words=f.read().split()
with open('output.txt','w') as f:
    for w in words:
        if len(w) > 1 and w==w[::-1]:
            f.write(w+'\n')
```

	Test	Input	Expected	Got	

✓	`with open('output.txt', 'r') as file:`
	`  text = file.read()`
	`  print(text)`
		input1.txt	madam
			arora
			malayalam
				madam
				arora
				malayalam
					✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **6**

Correct

Mark 1.00 out of 1.00

Write a Python program to append a new line at a specific position in a text file, shifting existing lines down.

Description:

1. **Input:**
   - A text file with multiple lines.
   - A line number to insert the new line at.
   - New content for the new line.
2. **Output:**
   - The updated file with the new line inserted at the specified position, shifting the existing lines down in file "output.txt".

Example:

- **Input File Content:**

  "Line one.
  Line two.
  Line three.
  Line four."
  3

  Inserted line..

Output:

Line one.
Line two.
Inserted line.
Line three.
Line four.

For example:

Test	Input	Result

`with open('output.txt', 'r') as file:`
`  text = file.read()`
`  print(text)`
	`input1.txt`
	`3`
	`Inserted line.`
		`Line one.`
		`Line two.`
		`Inserted line.`
		`Line three.`
		`Line four.`

Answer: (penalty regime: 0 %)

```
name=input()
n=int(input())
new=input()
with open(name) as f:
    lines=f.read().splitlines()
lines.insert(n-1,new)
with open('output.txt','w') as f:
    f.write('\n'.join(lines)+'\n')
```

	Test	Input	Expected	Got	

✓	`with open('output.txt', 'r') as file:`
	`  text = file.read()`
	`  print(text)`
		`input1.txt`
		`3`
		`Inserted line.`
			`Line one.`
			`Line two.`
			`Inserted line.`
			`Line three.`
			`Line four.`
				`Line one.`
				`Line two.`
				`Inserted line.`
				`Line three.`
				`Line four.`
					✓

✓	`with open('output.txt', 'r') as file:`
	`  text = file.read()`
	`  print(text)`
		`input2.txt`
		`4`
		`Inserted line D.`
			`Line A.`
			`Line B.`
			`Line C.`
			`Inserted line D.`
				`Line A.`
				`Line B.`
				`Line C.`
				`Inserted line D.`
					✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **7**

Correct

Mark 1.00 out of 1.00

Develop a Python program to copy the contents of one file to another file.

Description:

1. **Input:**
   - Source file and destination file names.
2. **Output:**
   - The content of the source file copied to the destination file.

For example:

Test	Input	Result

`with open('output1.txt', 'r') as file:`
`  text = file.read()`
`  print(text)`
	input1.txt
	output1.txt
		This is the source file.
		It contains multiple lines of text.
		Here is another line.

Answer: (penalty regime: 0 %)

```
src=input()
dst=input()
with open(src) as f:
    data=f.read()
with open(dst, 'w') as f:
    f.write(data)
```

	Test	Input	Expected	Got	

✓	`with open('output1.txt', 'r') as file:`
	`  text = file.read()`
	`  print(text)`
		input1.txt
		output1.txt
			This is the source file.
			It contains multiple lines of text.
			Here is another line.
				This is the source file.
				It contains multiple lines of text.
				Here is another line.
					✓

✓	`with open('output2.txt', 'r') as file:`
	`  text = file.read()`
	`  print(text)`
		input2.txt
		output2.txt
			Hello, world!
			Python programming is amazing.
			Let's copy this text to another file.
				Hello, world!
				Python programming is amazing.
				Let's copy this text to another file.
					✓

✓	`with open('output3.txt', 'r') as file:`
	`  text = file.read()`
	`  print(text)`
		input3.txt
		output3.txt
			Single line.	Single line.	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **8**

Correct

Mark 1.00 out of 1.00

Develop a Python program to read a specific line from a text file based on a given line number.

Description:

1. **Input:**
   - A text file with multiple lines.
   - A line number to read.
2. **Output:**
   - The content of the specified line.

input1.txt:

Line one.
Line two.
Line three.
Line four.

For example:

Input	Result
input1.txt 3	Line three.

Answer: (penalty regime: 0 %)

```
name=input()
n=int(input())
with open(name) as f:
    lines=f.read().splitlines()
if 1<=n<=len(lines):
    print(lines[n-1])
```

	Input	Expected	Got	
✓	input1.txt 3	Line three.	Line three.	✓
✓	input2.txt 3	Line C.	Line C.	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **9**

Correct

Mark 1.00 out of 1.00

Create a Python program to delete a specific line from a text file based on a given line number.

Description:

1. **Input:**
   - A text file with multiple lines.
   - A line number to delete.
2. **Output:**
   - The updated file with the specified line removed in file "output.txt".

Example:

- **Input File Content:**

  "Line one.
  Line two.
  Line three.
  Line four."
  2

  Updated line two.

Output:

Line one.
Line three.
Line four.

For example:

Test	Input	Result

`with open('output.txt', 'r') as file:`
`  text = file.read()`
`  print(text)`
	input1.txt
	2
		Line one.
		Line three.
		Line four.

Answer: (penalty regime: 0 %)

```
name=input()
n=int(input())
with open(name) as f:
    lines=f.read().splitlines()
if 1<=n<=len(lines):
    del lines[n-1]
with open('output.txt','w') as f:
    f.write('\n'.join(lines)+'\n')
```

	Test	Input	Expected	Got	

✓	`with open('output.txt', 'r') as file:`
	`  text = file.read()`
	`  print(text)`
		input1.txt
		2
			Line one.
			Line three.
			Line four.
				Line one.
				Line three.
				Line four.
					✓

✓	`with open('output.txt', 'r') as file:`
	`  text = file.read()`
	`  print(text)`
		input2.txt
		3
			Line A.
			Line B.
				Line A.
				Line B.
					✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **10**

Correct

Mark 1.00 out of 1.00

Develop a Python program to read a text file and count the total number of words in the file.

Description:

1. **Input:**
   - A text file containing several lines of text.
   - File name you should get as input.
2. **Output:**
   - The total number of words in the file.

For example:

Input	Result
input2.txt	Total words: 14
input3.txt	Total words: 15

Answer: (penalty regime: 0 %)

```
name=input()
with open(name) as f:
    words=f.read().split()
print("Total words:",len(words))
```

	Input	Expected	Got	
✓	input1.txt	Total words: 6	Total words: 6	✓
✓	input2.txt	Total words: 14	Total words: 14	✓
✓	input3.txt	Total words: 15	Total words: 15	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.